User Manual

for S32K3 PWM Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	31.03.2022	NXP RTD Team	Prepared for release RTD S32K3 2.0.0

Chapter 2

Introduction

- Supported Derivatives
- Overview
- About This Manual
- Acronyms and Definitions
- Reference List

This User Manual describes NXP Semiconductor AUTOSAR PWM for S32K3. AUTOSAR PWM driver configuration parameters and deviations from the specification are described in Driver chapter of this document. AUTOSAR PWM driver requirements and APIs are described in the AUTOSAR PWM driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k344_mapbga257
- s32k344_mqfp172
- s32k324_mapbga257
- s32k324_mqfp172
- s32k314_mapbga257
- s32k314_mqfp172
- s32k312_mqfp172
- s32k312_mqfp100
- s32k322_mqfp172
- s32k322_mqfp100
- s32k342_mqfp172
- s32k342_mqfp100
- s32k341_mqfp172
- s32k341_mqfp100

All of the above microcontroller devices are collectively named as S32K3.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Signifigant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of PWM Driver	AUTOSAR Release 4.4.0
2	Hardware Safety Manual	S32K3xx Safety Manual, Rev. 1, Sep 2021
3	S32K3XX Reference Manual	S32K3xx Reference Manual, Rev. 3, 10/2021
4	S32K3XX Data Sheet	S32K3xx Data Sheet, Rev. 3, 10/2021
5	S32K344 Errata	Mask Set Errata for Mask 0P55A/1P55A, Rev. 30/Sep/2021
6	S32K312 Errata	Mask Set Errata for Mask 0P09C, Rev. 10/Jan/2022
7	S32K342 Errata	Mask Set Errata for Mask 0P97C, Rev. 13/Dec/2021

Chapter 3

Driver

- Requirements
- Driver Design Summary
- Hardware Resources
- Deviations from Requirements
- Driver Limitations
- Driver usage and configuration tips
- Runtime errors
- Symbolic Names Disclaimer

3.1 Requirements

Requirements for this driver are detailed in the Autosar Driver Software Specification document (See Table Reference List).

3.2 Driver Design Summary

The driver provides functions for initialization and control of the microcontroller internal PWM stage (pulse width modulation). The PWM module generates pulses with variable pulse width. It allows the selection of the duty cycle and the signal period time. Each PWM channel is linked to a hardware channel capable of implementing PWM functionality, which belongs to the microcontroller.

The S32K3XX microcontroller contains three instances of Emios modules and one instance of FlexIO module.

The Emios module has the following major features in this release:

- Up to 24 channels chosen among Unified or Dedicated Channels, not necessarily numbered in a continuous sequence.

- Data registers of either 8-, 16-, 24-, or 32-bit width.
- Counter buses B, C, D, and E can be driven by Unified Channels 0, 8, 16, and 24, respectively.
- Counter bus A can be driven by Unified Channel 23.
- Counter bus F can be driven by a specified Unified Channel, defined by the system configuration.
- Each channel has its own timebase, alternative to the counter buses.
- Two global prescalers.
- One prescaler per channel (CP).
- State of the Unified Channels can be frozen for debug purposes.
- Output Pulse-Width and Frequency Modulation Buffered (OPWFMB).

The Flexio module has the following major features in this release:

- Up to 8 channels chosen among the Flexio timers.
- Up to 32 pins that can be chosen for any channel.
- Timer 8-bit High PWM Mode.
- Timer 8-bit Low PWM Mode.
- Pin output override mode to drive either HIGH or LOW signal on channel output.
- One prescaler per channel.
- State of the timers can be frozen for debug purposes.

3.3 Hardware Resources

Resources available on each derivative.

Derivatives	Pwm Module	Pwm Channel available
s32k312_mqfp172	Emios 0, Emios 1	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k312_mqfp100	Emios 0, Emios 1	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7

Derivatives	Pwm Module	Pwm Channel available
s32k314_mapbga257	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k314_mqfp100	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k314_mqfp172	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k324_mapbga257	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k324_mqfp172	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7
s32k344_mapbga257	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7

Derivatives	Pwm Module	Pwm Channel available
s32k344_mqfp172	Emios 0, Emios 1, Emios 2	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7, Channel 8, Channel 9, Channel 10, Channel 11, Channel 12, Channel 13, Channel 14, Channel 15, Channel 16, Channel 17, Channel 18, Channel 19, Channel 20, Channel 21, Channel 22, Channel 23
	Flexio	Channel 0, Channel 1, Channel 2, Channel 3, Channel 4, Channel 5, Channel 6, Channel 7

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR PWM Driver software specification in some places. The table below identifies the AUTOSAR requirements that are not implemented or out of scope for the PWM Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently or out of scope for the PWM driver.

Requirement	Status	Description	Notes
SWS_Pwm_00065	N/S	The Pwm SWS shall not define the code file structure.	Code file structure does always not define in the driver.
SWS_Pwm_70075	N/S	Pwm_Irq.c shall include Pwm.h.	Pwm_Irq.c is not needed. Autosar specific interrupt behaviour is implemented using a normal function placed in the Pwm.c file.
SWS_Pwm_00162	N/S	The Pwm Driver shall support synchronuous and asynchronous power state transitions, depending on the value of the configuration parameter PwmPower↩ StateAsynchTransitionMode.	Rejected: All PWM hardware do not support asynchronous power state transitions
SWS_Pwm_00164	N/S	In case the configuration parameter PwmPowerState↩ AsynchTransitionMode is set to TRUE, the preparation process shall continue in background after the relative API returns and its completion shall be notified by means of the configured callback.	Rejected: All PWM hardware do not support asynchronous power state transitions

Requirement	Status	Description	Notes
SWS_Pwm_00189	N/S	Service name: - Pwm_Main↩ _PowerTransitionManager - Syntax: - void Pwm_Main_↩ PowerTransitionManager(void) - Service ID[hex]: - 0x0d - Description: - This API is cyclically called and supervises the power state transitions, checking for the readiness of the module and issuing the callbacks IoHwAb_Pwm_↩ NotifyReadyForPowerState (see PwmPowerStateReadyCbkRef configuration parameter). - Available via: - SchM_Pwm.h -	Rejected: All PWM hardwares do not support asynchronous power state transitions
SWS_Pwm_00190	N/S	This API executes any non-immediate action needed to finalize a power state transition requested by Pwm_Prepare↩ PowerState().	Rejected: All PWM hardwares do not support asynchronous power state transitions
SWS_Pwm_00191	N/S	The rate of scheduling shall be defined by Pwm Main↩ SchedulePeriod and shall be variable, as the function only needs to be called if a transition has been requested.	Rejected: All PWM hardwares do not support asynchronous power state transitions
SWS_Pwm_00192	N/S	This API shall also issue callback notifications to the eventually registered users (IoHwAbs) as configured, only in case the asynch mode is chosen.	Rejected: All PWM hardwares do not support asynchronous power state transitions

Requirement	Status	Description	Notes
SWS_Pwm_00193	N/S	In case the PWM module is not initialized, this function shall simply return without any further elaboration. This is needed to avoid to elaborate uninitialized variables. No DET error shall be entered, because this condition can easily be verified during the startup phase (tasks started before the initialization is complete).Rationale: during the startup phase it can happen that the OS already schedules tasks, which call main functions, while some modules are not initialised yet. This is no real error condition, although need handling, i.e. returning without execution.Although the transition state monitoring functionality is mandatory, the implementation of this API is optional, meaning that if the HW allows for other ways to deliver notification and watch the transition state the implementation of this function can be skipped.	Rejected: All PWM hardwares do not support asynchronous power state transitions
SWS_Pwm_00198	N/S	Service name: - IoHwAb↩ _Pwm_NotifyReadyFor↩ PowerState - Syntax: - void IoHwAb_Pwm_NotifyReady↩ ForPowerState(void) - Service ID[hex]: - 0x60 - Sync/Async: - Synchronous - Reentrancy: - Non Reentrant - Parameters (in): - None - Parameters (inout): - None - Parameters (out): - None - Return value: - None - Description: - The API shall be invoked by the PWM Driver when the requested power state preparation for mode is completed. - Available via: - IoHwAb_Pwm.h -	Rejected: All PWM hardwares do not support asynchronous power state transitions

Requirement	Status	Description	Notes
SWS_Pwm_00199	N/S	In case the PWM Driver is configured to support power state management with asynchronous transitions, this API shall be called to signal completion of the power transition preparation phase to the IoHwAbs module.This is a callback, this API is to be implemented in the IoHwAbs component.	Rejected: All PWM hardwares do not support asynchronous power state transitions
SWS_Pwm_00153	N/S	These requirements are not applicable to this specification.	not a requirement
ECUC_Pwm_00143	N/S	Name - PwmPowerState↩ AsynchTransitionMode - Parent Container - PwmGeneral - Description - Enables / disables support of the PWM Driver to the asynchronous power state transition. - Multiplicity - 0..1 - Type - EcucBooleanParamDef - Default value - false - Post-Build Variant Multiplicity - false - Post-Build Variant Value - false - Multiplicity Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: localdependency↩ : This parameter shall only be configured if the parameter PwmLowPowerStatesSupport is set to true. -	Rejected: All PWM hardwares do not support asynchronous power state transitions
ECUC_Pwm_00144	N/S	Container Name - PwmPower↩ StateConfig - Description - Each instance of this parameter defines a power state and the callback to be called when this power state is reached. - Configuration Parameters -	Rejected: All PWM hardwares do not support asynchronous power state transitions

Requirement	Status	Description	Notes
ECUC_Pwm_00146	N/S	Name - PwmPowerState - Parent Container - PwmPower↩ StateConfig - Description - Each instance of this parameter describes a different power state supported by the PWM HW. It should be defined by the HW supplier and used by the P↩ WMDriver to reference specific HW configurations which set the PWM HW module in the referenced power state.At least the power mode corresponding to full power state shall be always configured. - Multiplicity - 1 - Type - EcucIntegerParamDef (Symbolic Name generated for this parameter) - Range - 0 .. 18446744073709551615 - - Default value - – - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope↩ : localdependency: This parameter shall only be configured if the parameter PwmLowPower↩ StatesSupport is set to true. -	Rejected: All PWM hardwares do not support asynchronous power state transitions
ECUC_Pwm_00145	N/S	Name - PwmPowerState↩ ReadyCbkRef - Parent Container - PwmPowerStateConfig - Description - Each instance of this parameter contains a reference to a power mode callback defined in a CDD or IoHwAbs component. - Multiplicity - 1 - Type - EcucFunctionNameDef - Default value - – - maxLength - – - minLength - – - regular↩ Expression - – - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: localdependency↩ : This parameter shall only be configured if the parameter PwmLowPowerStatesSupport is set to true. -	Rejected: All PWM hardwares do not support asynchronous power state transitions

Requirement	Status	Description	Notes
SWS_Pwm_CONSTR_00001	N/S	DRAFT: The ECUC partitions referenced by PwmKernel↩ EcucPartitionRef shall be a subset of the ECUC partitions referenced by PwmEcuc↩ PartitionRef.	Type IV Autosar multicore not implemented for current module (AAI-445), therefore Pwm↩ KernelEcucPartitionRef is not supported
SWS_Pwm_91003	N/S	Service name: - Pwm_Disable↩ Notification (draft) - Syntax: - void Pwm_DisableNotification(Pwm_ChannelType Channel↩ Number) - Service ID[hex]↩ : - 0x06 - Sync/Async: - Asynchronous - Reentrancy: - Reentrant for different channel numbers - Parameters (in): - ChannelNumber - Numeric identifier of the PWM - Parameters (inout): - None - Parameters (out): - None - Return value: - None - Description↩ : - Service to disable the P↩ WM signal edge notification.↩ Tags: atp.Status=draft - Available via: - Pwm.h -	Description specified as draft is not clear. Should be re-assesed on next ASR version
SWS_Pwm_91004	N/S	Service name: - Pwm_↩ EnableNotification (draft) - Syntax: - void Pwm_↩ EnableNotification(Pwm_↩ ChannelType ChannelNumber, Pwm_EdgeNotificationType Notification) - Service ID[hex]: - 0x07 - Sync/Async: - Asynchronous - Reentrancy: - Reentrant for different channel numbers - Parameters (in): - ChannelNumber - Numeric identifier of the PWM - Notification - Type of notification PWM_RISING_EDGE or PWM_FALLING_EDGE or PWM_BOTH_EDGES - Parameters (inout): - None - Parameters (out): - None - Return value: - None - Description: - Service to enable the PWM signal edge notification according to notification parameter.Tags: atp.Status=draft - Available via: - Pwm.h -	Description specified as draft is not clear. Should be re-assesed on next ASR version

Requirement	Status	Description	Notes
SWS_Pwm_91000	N/S	Service name: - Pwm_Set↩ DutyCycle (draft) - Syntax↩ : - void Pwm_SetDutyCycle(Pwm_ChannelType Channel↩ Number, uint16 DutyCycle) - Service ID[hex]: - 0x02 - Sync/Async: - Asynchronous - Reentrancy: - Reentrant for different channel numbers - Parameters (in): - Channel↩ Number - Numeric identifier of the PWM - DutyCycle - Min=0x0000 Max=0x8000 - Parameters (inout): - None - Parameters (out): - None - Return value: - None - Description↩ : - Service sets the duty cycle of the PWM channel.Tags: atp.↩ Status=draft - Available via: - Pwm.h -	Description specified as draft is not clear. Should be re-assesed on next ASR version
SWS_Pwm_91002	N/S	Service name: - Pwm_Set↩ OutputToIdle (draft) - Syntax: - void Pwm_SetOutputToIdle(Pwm_ChannelType Channel↩ Number) - Service ID[hex]↩ : - 0x04 - Sync/Async: - Asynchronous - Reentrancy: - Reentrant for different channel numbers - Parameters (in): - ChannelNumber - Numeric identifier of the PWM - Parameters (inout): - None - Parameters (out): - None - Return value↩ : - None - Description: - Service sets the PWM output to the configured Idle state.Tags↩ : atp.Status=draft - Available via: - Pwm.h -	Description specified as draft is not clear. Should be re-assesed on next ASR version

Requirement	Status	Description	Notes
SWS_Pwm_91001	N/S	Service name: - Pwm_↩ SetPeriodAndDuty (draft) - Syntax: - void Pwm_Set↩ PeriodAndDuty(Pwm_↩ ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle) - Service ID[hex]: - 0x03 - Sync/Async: - Asynchronous - Reentrancy: - Reentrant for different channel numbers - Parameters (in): - ChannelNumber - Numeric identifier of the PWM - Period - Period of the PWM signal - DutyCycle - Min=0x0000 Max=0x8000 - Parameters (inout): - None - Parameters (out): - None - Return value: - None - Description: - Service sets the period and the duty cycle of a PWM channelTags: atp.Status=draft - Available via: - Pwm.h -	Description specified as draft is not clear. Should be re-assesed on next ASR version
ECUC_Pwm_00150	N/S	Name - PwmKernelEcuc↩ PartitionRef - Parent Container - PwmGeneral - Description - Maps the PWM kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the PWM driver is mapped to.Tags: atp.Status=draft - Multiplicity - 0..1 - Type - Reference to [EcucPartition] - Post-Build Variant Multiplicity - true - Post-Build Variant Value - true - Multiplicity Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: ECU -	Type IV Autosar multicore not implemented for current module (AAI-445), therefore Pwm↩ KernelEcucPartitionRef is not supported

Requirement	Status	Description	Notes
SWS_Pwm_00175	N/S	The API shall report the DET error PWM_E_TRANSITIO↩ N_NOT_POSSIBLE in case the requested power state cannot be directly reached from the current power state.	Currently, Pwm driver has only 2 powe state (High power and Low Power) so this DET error will not happen
SWS_Pwm_00195	N/S	The API shall report the DET error PWM_E_TRANSITIO↩ N_NOT_POSSIBLE in case the requested power state cannot be directly reached from the current power state.	Currently, Pwm driver has only 2 powe state (High power and Low Power) so this DET error will not happen
SWS_Pwm_00188	N/S	The API shall report the DET error PWM_E_TRANSITIO↩ N_NOT_POSSIBLE in case the requested power state cannot be directly reached from the current power state.All asynchronous operation needed to reach the target power state can be executed in background in the context of Pwm_Main_↩ PowerTransitionManager.	s

3.5 Driver Limitations

None.

3.6 Driver usage and configuration tips

In order for the Pwm driver to function without problems the following steps must be taken:

- Initialize the MCU driver with the desired clock configuration
- Initialize the MCL driver with a valid Emios and Flexio instance configuration.
- Initialize the PORT driver with a desired pin configuration.
- Initialize the PWM driver.

3.7 Runtime errors

None.

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

#define <Mip>Conf_<Container_ShortName>_<Container_ID>

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module Pwm
  - Container PwmChannelConfigSet
    - Container PwmChannel
      - Parameter PwmChannelId
      - Parameter PwmChannelClass
      - Parameter PwmPeriodInTicks
      - Parameter PwmPeriodDefault
      - Parameter PwmDutycycleDefault
      - Parameter PwmPolarity
      - Parameter PwmIdleState
      - Parameter PwmNotification
      - Reference PwmChannelEcucPartitionRef
      - Reference PwmHwChannel
      - Reference PwmMcuClockReferencePoint
    - Container PwmEmios
      - Parameter PwmHwInstance
      - Container PwmEmiosChannels
      - Parameter EmiosChId
      - Parameter EmiosChMode
      - Parameter EmiosChFlagGeneration
      - Parameter EmiosChCounterBus
      - Parameter EmiosChFreeze
      - Parameter EmiosChOutputDisable
      - Parameter EmiosChPrescaler
      - Parameter EmiosChPrescalerAlternate
      - Parameter EmiosChPrescalerSource
      - Parameter EmiosChPolarity
      - Parameter EmiosChInterrupt
      - Parameter EmiosChDutyCycle
      - Parameter EmiosChPeriod

- Parameter EmiosChPhaseShift
- Parameter EmiosChTrigger
- Parameter EmiosChDeadtime
- Reference PwmEmiosBusRef
- Container EmiosChIrqCallback
- Parameter EmiosChIrqFunctionCallback
- Parameter EmiosChIrqParameterCallback

* Container PwmFlexio
  - Parameter PwmHwInstance
  - Container PwmFlexioChannels
  - Parameter FlexioChId
  - Parameter FlexioPinId
  - Parameter FlexioChPrescaler
  - Parameter FlexioChPrescalerAlternate
  - Parameter FlexioChDutyCycle
  - Parameter FlexioChPeriod
  - Parameter FlexioChPolarity
  - Parameter FlexioChInterrupt
  - Reference FlexioMclChRef
  - Container FlexioChIrqCallback
  - Parameter FlexioChIrqFunctionCallback
  - Parameter FlexioChIrqParameterCallback

– Container PwmGeneral
  * Parameter PwmMulticoreEnabled
  * Parameter PwmDevErrorDetect
  * Parameter PwmDutycycleUpdatedEndperiod
  * Parameter PwmPeriodUpdatedEndperiod
  * Parameter PwmNotificationSupported
  * Parameter PwmEnableUserModeSupport
  * Parameter PwmLowPowerStatesSupport
  * Parameter PwmPowerStateAsynchTransitionMode
  * Parameter PwmEnableDualClockMode
  * Parameter PwmMultiChannelSync
  * Parameter PwmIndex
  * Reference PwmEcucPartitionRef
  * Reference PwmKernelEcucPartitionRef
  * Container PwmPowerStateConfig
    - Parameter PwmPowerState
    - Parameter PwmPowerStateReadyCbkRef

– Container PwmConfigurationOfOptApiServices
  * Parameter PwmDeInitApi

    * Parameter PwmGetOutputState
    * Parameter PwmSetDutyCycle
    * Parameter PwmSetOutputToIdle
    * Parameter PwmSetPeriodAndDuty
    * Parameter PwmVersionInfoApi
    * Parameter PwmGetChannelStateApi
    * Parameter PwmSetDutyCycle_NoUpdate
    * Parameter PwmSetPeriodAndDuty_NoUpdate
    * Parameter PwmSetPhaseShift
    * Parameter PwmSetPhaseShift_NoUpdate
    * Parameter PwmSetDutyPhaseShift
    * Parameter PwmSetChannelDeadTime
    * Parameter PwmSetCounterBusApi
    * Parameter PwmSetChannelOutputApi
    * Parameter PwmSetTriggerDelayApi
  – Container CommonPublishedInformation
    * Parameter ArReleaseMajorVersion
    * Parameter ArReleaseMinorVersion
    * Parameter ArReleaseRevisionVersion
    * Parameter ModuleId
    * Parameter SwMajorVersion
    * Parameter SwMinorVersion
    * Parameter SwPatchVersion
    * Parameter VendorApiInfix
    * Parameter VendorId

4.1 Module Pwm

Configuration of Pwm (Pulse Width Modulation) module.

Included containers:

- PwmChannelConfigSet
- PwmGeneral
- PwmConfigurationOfOptApiServices
- CommonPublishedInformation

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container PwmChannelConfigSet

Container contains the channel configuration parameter of the Pwm driver.

Included subcontainers:

- PwmChannel
- PwmEmios
- PwmFlexio

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Container PwmChannel

Configuration of an individual Pwm channel.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.4 Parameter PwmChannelId

Channel ID of the Pwm channel. This value will be assigned to the symbolic name derived of the PwmChannel container short name.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967295
min	0

4.5 Parameter PwmChannelClass

Class of Pwm Channel.

PWM_FIXED_PERIOD - Period of the channel will not be changed.

PWM_FIXED_PERIOD_SHIFTED - Period of the channel will not be changed, and support with phase shift feature.

PWM_VARIABLE_PERIOD - Period of the channel can be changed.

<note>

Due to Ftm hardware specific feature that the counter register and period register are shared common for all

channels in one Ftm instance, therefore if current channel is configured PWM_FIXED_PERIOD class, but another

channel in the same Ftm instance is configured PWM_VARIABLE_PERIOD, then when that channel call function to

change period, current channel will change period, too.

</note>

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PWM_FIXED_PERIOD
literals	['PWM_FIXED_PERIOD', 'PWM_FIXED_PERIOD_SHIFTED', 'PWM_↩ VARIABLE_PERIOD']

4.6 Parameter PwmPeriodInTicks

Check this option to configure Default Period unit in ticks, or uncheck this to configure Default Period unit in seconds.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.7 Parameter PwmPeriodDefault

Default period value of Pwm channel at initialization.

The measure unit are in ticks (if Period In Ticks checked), or in seconds (if unchecked)

Valid range: [0, 65534]

<note>

The maximum period tick is 65534 (instead of maximum value of period register 65535) in order to achieve perfect 0 or 100% duty cycle.

All channels which are in the same Ftm instance with current channel must have the same Default Period value, due to Ftm hardware specific feature that the period register is shared common for all channels in one Ftm instance.

</note>

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1.0
max	65534.0
min	0.0

4.8 Parameter PwmDutycycleDefault

Default value for duty cycle of Pwm channel at initialization.

0 represents for 0% duty cycle

16384 (0x4000) represents for 50% duty cycle

32768 (0x8000) represents for 100% duty cycle

Valid value: [0,32768]

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	16384
max	32768
min	0

4.9 Parameter PwmPolarity

Define the polarity of Pwm channel at initialization.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PWM_HIGH
literals	['PWM_HIGH', 'PWM_LOW']

4.10 Parameter PwmIdleState

Define Pwm channel state when the output is set to idle.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PWM_LOW
literals	['PWM_HIGH', 'PWM_LOW']

4.11 Parameter PwmNotification

User callback notification function.

This option is only activated when PwmGeneral/PwmNotificationSupported is checked.

<note>

Use NULL_PTR without any quotes to determine no notification function is used.

If the string is different from above, it will be used as the notification function name.

Notification does not apply to channel that its alignment type is PWM_CENTER_ALIGNED.

</note>

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.12 Reference PwmChannelEcucPartitionRef

Maps a Pwm channel to zero or multiple ECUC partitions to limit the access to this channel group.

The ECUC partitions referenced are a subset of the ECUC partitions where the Pwm driver is mapped to.

When users choose ENABLE multicore feature by checking PwmMulticoreEnabled option, this will force to configure at least 1 ECUC partition in this list that is referenced from ECUC module; OR when DISABLE multicore feature, user have to remove all ECUC partitions in this list.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.13 Reference PwmHwChannel

Select the hw channel on which the functionality of the current PWM channel will be implemented.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/TS_T40D34M20I0R0/Pwm/PwmChannelConfigSet/PwmEmios/Pwm↩ EmiosChannels', '/TS_T40D34M20I0R0/Pwm/PwmChannelConfigSet/Pwm↩ Flexio/PwmFlexioChannels']

4.14 Reference PwmMcuClockReferencePoint

Reference to the clock source configuration, which is set in the MCU driver configuration.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSetting↩ Config/McuClockReferencePoint

4.15 Container PwmEmios

Configuration of an Emios module available on the platform.

Included subcontainers:

- PwmEmiosChannels

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	3
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.16 Parameter PwmHwInstance

Select the hardware Emios module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Emios_0
literals	['Emios_0', 'Emios_1', 'Emios_2']

4.17 Container PwmEmiosChannels

List of Emios channels available on the platform.

Included subcontainers:

- EmiosChIrqCallback

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	24
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.18 Parameter EmiosChId

Select one of the Emios channels available on the platform.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CH_0
literals	['CH_0', 'CH_1', 'CH_2', 'CH_3', 'CH_4', 'CH_5', 'CH_6', 'CH_7', 'CH_↩ 8', 'CH_9', 'CH_10', 'CH_11', 'CH_12', 'CH_13', 'CH_14', 'CH_15', 'CH_16', 'CH_17', 'CH_18', 'CH_19', 'CH_20', 'CH_21', 'CH_22', 'CH_23']

4.19 Parameter EmiosChMode

Select the mode for the Emios channel:

OPWFMB: Variable Period and Variable Duty Cycle. Bus: Internal only

OPWMB: Fixed Period, Variable Duty Cycle Bus: External bus only

OPWMCB: Variable Period and Variable Duty Cycle with dead time insertion Bus: External bus only

OPWMT: Fixed Period, Variable Duty Cycle with trigger generation Bus: External bus only

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_MODE_OPWFMB
literals	['EMIOS_PWM_IP_MODE_OPWFMB', 'EMIOS_PWM_IP_MODE_OP↩ WMCB_TRAIL_EDGE', 'EMIOS_PWM_IP_MODE_OPWMCB_LEAD_↩ EDGE', 'EMIOS_PWM_IP_MODE_OPWMB', 'EMIOS_PWM_IP_MOD↩ E_OPWMT', 'EMIOS_PWM_IP_MODE_DAOC']

4.20 Parameter EmiosChFlagGeneration

Select when the flag will be set.

FLAG: FLAG will be set on trailing edge.

FLAG_BOTH: FLAG will be set on leading edge and trailing edge.

For OPWMT mode Flag will only be set when a trigger is generated.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Trailing_Edge
literals	['Trailing_Edge', 'Both_Trailing_and_Leading_Edge']

4.21 Parameter EmiosChCounterBus

Select the counter bus used by the channel.

EMIOS_PWM_IP_BUS_A - counter bus A, available for all channels

EMIOS_PWM_IP_BUS_F - counter bus F, available for all channels

EMIOS_PWM_IP_BUS_BCDE - counter bus B, C, D or E specific to each channel

Bus B is controlled by channel 0 and is available for channels 0->7

Bus C is controlled by channel 8 and is available for channels 8->15

Bus D is controlled by channel 16 and is available for channels 16->23

Bus E is controlled by channel 24 and is available for channels 24->31

EMIOS_PWM_IP_BUS_INTERNAL - unified channel internal counter

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_BUS_A
literals	['EMIOS_PWM_IP_BUS_A', 'EMIOS_PWM_IP_BUS_BCDE', 'EMIOS_↩ PWM_IP_BUS_F', 'EMIOS_PWM_IP_BUS_INTERNAL']

4.22 Parameter EmiosChFreeze

This parameter controls Freeze Enable bit (FREN), if set and validated by FRZ bit in EMIOS_MCR register, freezes all registers value of period used for initialization.

True - Freeze channel registers values.

False - Normal operation.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.23 Parameter EmiosChOutputDisable

Select one of the four output disable input signals.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_OUTPUT_DISABLE_NONE
literals	['EMIOS_PWM_IP_OUTPUT_DISABLE_NONE', 'EMIOS_PWM_IP_O↩ UTPUT_DISABLE_0', 'EMIOS_PWM_IP_OUTPUT_DISABLE_1', 'EMI↩ OS_PWM_IP_OUTPUT_DISABLE_2', 'EMIOS_PWM_IP_OUTPUT_D↩ ISABLE_3']

4.24 Parameter EmiosChPrescaler

Select clock prescaler used for this Emios channel. The internal counter must be selected for this setting to take any effect.

Will only have an effect in OPWFMB and OPWMCB modes.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_CLOCK_DIV_1
literals	['EMIOS_PWM_IP_CLOCK_NONE', 'EMIOS_PWM_IP_CLOCK_DIV_1', 'EMIOS_PWM_IP_CLOCK_DIV_2', 'EMIOS_PWM_IP_CLOCK_DIV_3', 'EMIOS_PWM_IP_CLOCK_DIV_4', 'EMIOS_PWM_IP_CLOCK_DIV_5', 'EMIOS_PWM_IP_CLOCK_DIV_6', 'EMIOS_PWM_IP_CLOCK_DIV_7', 'EMIOS_PWM_IP_CLOCK_DIV_8', 'EMIOS_PWM_IP_CLOCK_DIV_↩ 9', 'EMIOS_PWM_IP_CLOCK_DIV_10', 'EMIOS_PWM_IP_CLOCK_D↩ IV_11', 'EMIOS_PWM_IP_CLOCK_DIV_12', 'EMIOS_PWM_IP_CLOC↩ K_DIV_13', 'EMIOS_PWM_IP_CLOCK_DIV_14', 'EMIOS_PWM_IP_C↩ LOCK_DIV_15', 'EMIOS_PWM_IP_CLOCK_DIV_16']

4.25 Parameter EmiosChPrescalerAlternate

Select alternative clock prescaler used for this Emios channel. The internal counter must be selected for this setting to take any effect.

Will only have an effect in OPWFMB and OPWMCB modes.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_CLOCK_DIV_1

Property	Value
literals	['EMIOS_PWM_IP_CLOCK_NONE', 'EMIOS_PWM_IP_CLOCK_DIV_1', 'EMIOS_PWM_IP_CLOCK_DIV_2', 'EMIOS_PWM_IP_CLOCK_DIV_3', 'EMIOS_PWM_IP_CLOCK_DIV_4', 'EMIOS_PWM_IP_CLOCK_DIV_5', 'EMIOS_PWM_IP_CLOCK_DIV_6', 'EMIOS_PWM_IP_CLOCK_DIV_7', 'EMIOS_PWM_IP_CLOCK_DIV_8', 'EMIOS_PWM_IP_CLOCK_DIV_↩ 9', 'EMIOS_PWM_IP_CLOCK_DIV_10', 'EMIOS_PWM_IP_CLOCK_D↩ IV_11', 'EMIOS_PWM_IP_CLOCK_DIV_12', 'EMIOS_PWM_IP_CLOC↩ K_DIV_13', 'EMIOS_PWM_IP_CLOCK_DIV_14', 'EMIOS_PWM_IP_C↩ LOCK_DIV_15', 'EMIOS_PWM_IP_CLOCK_DIV_16']

4.26 Parameter EmiosChPrescalerSource

Select clock source for the internal prescaler used for this Emios channel.

Will only have an effect in OPWFMB and OPWMCB modes.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_PS_SRC_MODULE_CLOCK
literals	['EMIOS_PWM_IP_PS_SRC_PRESCALED_CLOCK', 'EMIOS_PWM_I↩ P_PS_SRC_MODULE_CLOCK']

4.27 Parameter EmiosChPolarity

Define the output polarity of the channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_ACTIVE_HIGH
literals	['EMIOS_PWM_IP_ACTIVE_HIGH', 'EMIOS_PWM_IP_ACTIVE_LOW']

4.28 Parameter EmiosChInterrupt

Define what happens when a flag event is generated.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	EMIOS_PWM_IP_NOTIFICATION_DISABLED
literals	['EMIOS_PWM_IP_NOTIFICATION_DISABLED', 'EMIOS_PWM_IP_I↩ NTERRUPT_REQUEST', 'EMIOS_PWM_IP_DMA_REQUEST']

4.29 Parameter EmiosChDutyCycle

Value for duty cycle used for initialization.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	32767
max	65534
min	0

4.30 Parameter EmiosChPeriod

Period value used at initialization.

NOTE: Only used in OPWFMB mode.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	65534
max	65534
min	1

4.31 Parameter EmiosChPhaseShift

Phase Shift (in ticks) of the PWM output.

Please note that the counter will always start from 1, and the Phase Shift value is incremented by 1 in the code.

The Phase Shift value must be less than the period of the channel used as reference or else a DET error will be generated.

NOTE: The Phase Shift parameter can only be used in OPWMB or OPWMT mode.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65534
min	0

4.32 Parameter EmiosChTrigger

Select input count source (clock) used for this Emios channelDelay (in ticks) for generating the trigger event.

Please note that the counter will always start from 1, and the Trigger value is incremented by 1 in the code, then the value is updated into Alternate A register.

The programmed value in the A2 register must be less or equal to the value of the channel period used as reference or else the CTU triggers will not be generated.

NOTE: This parameter is used only in OPWMT mode.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65534
min	0

4.33 Parameter EmiosChDeadtime

Deadtime parameter controls the deadtime during transitions of the PWM output.

NOTE: This parameter is used only for OPWMCB mode.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65534
min	0

4.34 Reference PwmEmiosBusRef

Select the masterbus channel on which the functionality of the current emios channel will be implemented.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/TS_T40D34M20I0R0/Mcl/MclConfig/EmiosCommon/EmiosMclMasterBus']

4.35 Container EmiosChIrqCallback

Configure Notfication function and parameter for interrupt handler callback.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.36 Parameter EmiosChIrqFunctionCallback

User callback function.

NOTE: Use NULL_PTR without any quotes. If the used string is different from NULL_PTR it will be used as the configured function name.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.37 Parameter EmiosChIrqParameterCallback

User callback parameter.

NOTE: Use NULL_PTR without any quotes. If the used string is different from NULL_PTR it will be used as the configured parameter name.

Property	Value
type	ECUC-FUNCTION-NAME-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.38 Container PwmFlexio

Configuration of an Flexio module available on the platform.

Included subcontainers:

- PwmFlexioChannels

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.39 Parameter PwmHwInstance

Select the hardware Flexio module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Flexio_0
literals	['Flexio_0']

4.40 Container PwmFlexioChannels

List of Flexio channels available on the platform.

Included subcontainers:

- FlexioChIrqCallback

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	8
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.41 Parameter FlexioChId

Select one of the Flexio channels available on the platform.

NOTE: This also selects the used timer.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CH_0
literals	['CH_0', 'CH_1', 'CH_2', 'CH_3', 'CH_4', 'CH_5', 'CH_6', 'CH_7']

4.42 Parameter FlexioPinId

Select one of the Flexio Pins available on the platform.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PIN_0
literals	['PIN_0', 'PIN_1', 'PIN_2', 'PIN_3', 'PIN_4', 'PIN_5', 'PIN_6', 'PIN_7', 'P↩IN_8', 'PIN_9', 'PIN_10', 'PIN_11', 'PIN_12', 'PIN_13', 'PIN_14', 'PIN_15', 'PIN_16', 'PIN_17', 'PIN_18', 'PIN_19', 'PIN_20', 'PIN_21', 'PIN_22', 'PIN↩_23', 'PIN_24', 'PIN_25', 'PIN_26', 'PIN_27', 'PIN_28', 'PIN_29', 'PIN_30', 'PIN_31']

4.43 Parameter FlexioChPrescaler

Select clock prescaler used for this Flexio channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FLEXIO_PWM_IP_CLK_DIV_1
literals	['FLEXIO_PWM_IP_CLK_DIV_1', 'FLEXIO_PWM_IP_CLK_DIV_16', 'FLEXIO_PWM_IP_CLK_DIV_256']

4.44 Parameter FlexioChPrescalerAlternate

Select alternative clock prescaler used for this Flexio channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FLEXIO_PWM_IP_CLK_DIV_1
literals	['FLEXIO_PWM_IP_CLK_DIV_1', 'FLEXIO_PWM_IP_CLK_DIV_16', 'FLEXIO_PWM_IP_CLK_DIV_256']

4.45 Parameter FlexioChDutyCycle

Value for duty cycle used for initialization.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	256
min	0

4.46 Parameter FlexioChPeriod

Period value used at initialization.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	512
min	0

4.47 Parameter FlexioChPolarity

Define the output polarity of the channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FLEXIO_PWM_IP_ACTIVE_HIGH
literals	['FLEXIO_PWM_IP_ACTIVE_HIGH', 'FLEXIO_PWM_IP_ACTIVE_LOW']

4.48 Parameter FlexioChInterrupt

Define what happens when a flag event is generated.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FLEXIO_PWM_IP_IRQ_DISABLED
literals	['FLEXIO_PWM_IP_IRQ_DISABLED', 'FLEXIO_PWM_IP_IRQ_ON_↩ RISING_EDGE', 'FLEXIO_PWM_IP_IRQ_ON_FALLING_EDGE', 'FLE↩ XIO_PWM_IP_IRQ_ON_BOTH_EDGES', 'FLEXIO_PWM_IP_IRQ_O↩ N_PERIOD_END']

4.49 Reference FlexioMclChRef

Select the Flexio MCL hw channel on which the the current PWM channel will be implemented.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D34M20I0R0/Mcl/MclConfig/FlexioCommon/FlexioMclLogicChannels

4.50 Container FlexioChIrqCallback

Configure Notification function and parameter for interrupt handler callback.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.51 Parameter FlexioChIrqFunctionCallback

User callback function.

NOTE: Use NULL_PTR without any quotes. If the used string is different from NULL_PTR it will be used as the configured function name.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.52 Parameter FlexioChIrqParameterCallback

User callback parameter.

NOTE: Use NULL_PTR without any quotes. If the used string is different from NULL_PTR it will be used as the configured parameter name.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.53 Container PwmGeneral

Container used for storing the general configuration of Pwm.

Included subcontainers:

- PwmPowerStateConfig

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.54 Parameter PwmMulticoreEnabled

Switch to enable/disable multicore feature.

User can choose ENABLE multicore feature by checking this option, this will force to configure at least 1 ECUC partition in

PwmChannelEcucPartitionRef, and each Pwm channel in PwmChannel to configure at least 1 ECUC partition reference

in PwmChannelEcucPartitionRef container to fulfill generating code condition; OR uncheck this option to DISABLE

multicore feature, performing this action will force user to remove all ECUC partition reference in every Pwm channels contained

in PwmChannel and in PwmChannelEcucPartitionRef.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.55 Parameter PwmDevErrorDetect

Switch to enable/disable the development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.56 Parameter PwmDutycycleUpdatedEndperiod

Switch to enable the update of the duty cycle parameter at the end of the current period.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.57 Parameter PwmPeriodUpdatedEndperiod

Switch to enable the update of the period parameter at the end of the current period.

<note>In current implementation, this option is locked and always enable by default.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.58 Parameter PwmNotificationSupported

Switch to indicate that the notifications are supported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.59 Parameter PwmEnableUserModeSupport

If enabled, the Pwm module will adapt to run from User Mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.60 Parameter PwmLowPowerStatesSupport

If enabled, hardware offers low power mode and adds all power state management related APIs

(Pwm_SetPowerState, Pwm_GetCurrentPowerState, Pwm_GetTargetPowerState,

Pwm_PreparePowerState, Pwm_Main_PowerTransitionManager),

indicating if the hardware offers low power state management.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.61 Parameter PwmPowerStateAsynchTransitionMode

Enable the support of the Pwm driver to the asynchronous power state transition.

<note> This feature is not supported and is rejected, as all hardware modules do not support asynchronous power state transitions.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.62 Parameter PwmEnableDualClockMode

Switch to enable/disable dual clock mode feature, which will add/remove the service Pwm_SetClockMode() from the code.

This feature is used when the prescaler value needs to be changed to maintain same period at different frequency.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.63 Parameter PwmMultiChannelSync

Enable the update synchronous feature for several channels.

This option will activate the use of PwmSetDutyCycle_NoUpdate, PwmSetPeriodAndDuty_NoUpdate and PwmSetPhaseShift_NoUpdate or PwmSetDutyPhaseShift.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.64 Parameter PwmIndex

Specify the InstanceId of this module instance. If only one instance is present it shall have the ID 0.

<note>In current implementation, this feature is not used.</note>

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967295
min	0

4.65 Reference PwmEcucPartitionRef

Maps the Pwm driver to zero or multiple ECUC partitions to make the driver API available in the according partition.

Depending on the addressed timer resource the interfaces operate as follows.

When users choose ENABLE multicore feature by checking PwmMulticoreEnabled option, this will force to configure at least 1 ECUC partition in this list that is referenced from ECUC module; OR when DISABLE multicore feature, user have to remove all ECUC partitions in this list.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.66 Reference PwmKernelEcucPartitionRef

Maps the Pwm kernel to zero or one ECUC partitions to assign the driver kernel to a certain core.

The ECUC partition referenced is a subset of the ECUC partitions where the Pwm driver is mapped to.

<note>This feature is not supported and is rejected.</note>

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.67 Container PwmPowerStateConfig

Each instance of this parameter defines a power state and the callback to be called when this power state is reached.

<note>This feature is not supported and is rejected, as all hardware modules do not support asynchronous power state transitions.</note>

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.68 Parameter PwmPowerState

Each instance of this parameter describes a different power state supported by the Pwm hardware.

It should be defined by the hardware supplier and used by the Pwm Driver to reference specific HW configurations which set the Pwm HW module in the referenced power state.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	9223372036854775807
min	0

4.69 Parameter PwmPowerStateReadyCbkRef

Each instance of this parameter contains a reference to a power mode callback defined in a CDD or IoHwAbs component.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	NULL_PTR

4.70 Container PwmConfigurationOfOptApiServices

Container used for storing the configuration of all optional API's.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.71 Parameter PwmDeInitApi

Add/remove the service Pwm_DeInit() from the code.

This option will toggle PWM_DE_INIT_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.72 Parameter PwmGetOutputState

Add/remove the service Pwm_GetOutputState() from the code.

This option will toggle PWM_GET_OUTPUT_STATE_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.73 Parameter PwmSetDutyCycle

Add/remove the service Pwm_SetDutyCycle() from the code.

This option will toggle PWM_SET_DUTY_CYCLE_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.74 Parameter PwmSetOutputToIdle

Add/remove the service Pwm_SetOutputToIdle() from the code.

This option will toggle PWM_SET_OUTPUT_TO_IDLE_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.75 Parameter PwmSetPeriodAndDuty

Add/remove the service Pwm_SetPeriodAndDuty() from the code.

This option will toggle PWM_SET_PERIOD_AND_DUTY_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.76 Parameter PwmVersionInfoApi

Add/remove the service Pwm_GetVersionInfo() from the code.

This option will toggle PWM_VERSION_INFO_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.77 Parameter PwmGetChannelStateApi

Add/remove the service Pwm_GetChannelState() from the code.

This option will toggle PWM_GET_CHANNEL_STATE_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.78 Parameter PwmSetDutyCycle_NoUpdate

Add/remove the service Pwm_SetDutyCycle_NoUpdate() from the code.

This option will toggle PWM_SET_DUTY_CYCLE_NO_UPDATE_API define macro in Pwm_Cfg.h file.

<note>This option is activated only when PwmGeneral/PwmMultiChannelSync is enabled.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.79 Parameter PwmSetPeriodAndDuty_NoUpdate

Add/remove the service Pwm_SetPeriodAndDuty_NoUpdate() from the code.

This option will toggle PWM_SET_PERIOD_AND_DUTY_NO_UPDATE_API define macro in Pwm_Cfg.h file.

<note>This option is activated only when PwmGeneral/PwmMultiChannelSync is enabled.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.80 Parameter PwmSetPhaseShift

Add/remove the service Pwm_SetPhaseShift() from the code.

This option will toggle PWM_SET_PHASE_SHIFT_API define macro in Pwm_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.81 Parameter PwmSetPhaseShift_NoUpdate

Add/remove the service Pwm_SetPhaseShift_NoUpdate() from the code.

This option will toggle PWM_SET_PHASE_SHIFT_NO_UPDATE_API define macro in Pwm_Cfg.h file.

<note>This option is activated only when PwmGeneral/PwmMultiChannelSync is enabled.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.82 Parameter PwmSetDutyPhaseShift

Add/remove the service Pwm_SetDutyPhaseShift() from the code.

This option will toggle PWM_SET_DUTY_PHASE_SHIFT_API define macro in Pwm_Cfg.h file.

<note>This option is activated only when PwmGeneral/PwmMultiChannelSync is enabled.</note>

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.83 Parameter PwmSetChannelDeadTime

Add/remove the service Pwm_SetChannelDeadTime() from the code.

This option will toggle PWM_SETCHANNELDEADTIME_API define macro.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.84 Parameter PwmSetCounterBusApi

Switch to indicate that the Pwm_SetCounterBus is supported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.85 Parameter PwmSetChannelOutputApi

Switch to indicate that the Pwm_SetChannelOutput is supported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.86 Parameter PwmSetTriggerDelayApi

Switch to indicate that the PwmSetTriggerDelay is supported.

This function is called when the prescaler value needs to be change to maintain same period at different frequency.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.87 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.88 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.89 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.90 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.91 Parameter ModuleId

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	121
max	121
min	121

4.92 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.93 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.94 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.95 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>_>VendorId>_<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	

4.96 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

Chapter 5

Module Index

5.1 Software Specification

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Chapter 6

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6.1 Data Structures

Here are the data structures with brief descriptions:

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Chapter 7

Module Documentation

7.1 Emios Pwm IPL

7.1.1 Detailed Description

Function Reference

- void Emios_Pwm_Ip_InitChannel (uint8 Instance, Emios_Pwm_Ip_ChannelConfigType const *UserChCfg)

  Initialize PWM Mode.
- void Emios_Pwm_Ip_DeInitChannel (uint8 Instance, uint8 Channel)

  Reset eMIOS Channel to GPIO mode (reset default)
- void Emios_Pwm_Ip_ForceMatchLeadingEdge (uint8 Instance, uint8 Channel, boolean Enable)

  Allow the software to force the output flip-flop to the level corresponding to a match on leading edge. The FLAG bit is not set.
- void Emios_Pwm_Ip_ForceMatchTrailingEdge (uint8 Instance, uint8 Channel, boolean Enable)

  Allow the software to force the output flip-flop to the level corresponding to a match on trailing edge. The FLAG bit is not set.
- uint16 Emios_Pwm_Ip_GetPeriod (uint8 Instance, uint8 Channel)

  Get Period value in PWM mode.
- void Emios_Pwm_Ip_SetPeriod (uint8 Instance, uint8 Channel, uint16 NewPeriod)

  Set new Period value in PWM mode.
- uint16 Emios_Pwm_Ip_GetDutyCycle (uint8 Instance, uint8 Channel)

  Get Duty Cycle value in PWM mode.
- Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_SetDutyCycle (uint8 Instance, uint8 Channel, uint16 New↩ DutyCycle)

  Set new Duty Cycle value in PWM mode.
- uint16 Emios_Pwm_Ip_GetPhaseShift (uint8 Instance, uint8 Channel)

  Get Leading Edge Placement value in PWM mode.
- Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_SetPhaseShift (uint8 Instance, uint8 Channel, uint16 Phase↩ Shift)

  Set new Leading edge placement value in PWM mode.
- uint16 Emios_Pwm_Ip_GetDeadTime (uint8 Instance, uint8 Channel)

  Get dead time value in PWM mode.

- void Emios_Pwm_Ip_SetDeadTime (uint8 Instance, uint8 Channel, uint16 NewDeadTime)

  Set new dead time value in PWM mode.
- uint32 Emios_Pwm_Ip_GetTriggerPlacement (uint8 Instance, uint8 Channel)

  Get Trigger Placement value in PWM mode.
- void Emios_Pwm_Ip_SetTriggerPlacement (uint8 Instance, uint8 Channel, uint32 NewTriggerPlacement)

  Set new Trigger Placement value in PWM mode.
- Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_ChannelEnterDebugMode (uint8 Instance, uint8 Channel)

  Set a Channel enters freeze state, should be setting EMIOS_AllowEnterDebugMode first.
- void Emios_Pwm_Ip_ChannelStopDebugMode (uint8 Instance, uint8 Channel)

  Release a Channel from freeze state.
- Emios_Pwm_Ip_InterruptType Emios_Pwm_Ip_GetFlagRequest (uint8 Instance, uint8 Channel)

  Get the Unified Channel FLAG event generated. Interrupt or DMA request.
- void Emios_Pwm_Ip_SetFlagRequest (uint8 Instance, uint8 Channel, Emios_Pwm_Ip_InterruptType Event)

  Allow the Unified Channel FLAG bit to generate an interrupt signal or a DMA request signal.
- Emios_Pwm_Ip_OutputStateType Emios_Pwm_Ip_GetOutputState (uint8 Instance, uint8 Channel)

  Get the Unified Channel output pin logic level.
- void Emios_Pwm_Ip_SetOutputState (uint8 Instance, uint8 Channel, Emios_Pwm_Ip_OutputStateType OutputState)

  Set the state of output pin.
- void Emios_Pwm_Ip_SetOutputToNormal (uint8 Instance, uint8 Channel, uint16 DutyPercent, Emios_↩ Pwm_Ip_PolarityType Polarity, Emios_Pwm_Ip_PwmModeType Mode)

  Set the polarity and mode for current Channel as normal.
- Emios_Pwm_Ip_PwmModeType Emios_Pwm_Ip_GetChannelMode (uint8 Instance, uint8 Channel)

  Get mode of operation of the Unified Channel.
- uint8 Emios_Pwm_Ip_GetMasterBusChannel (uint8 Instance, uint8 Channel)

  Get master bus Channel.
- void Emios_Pwm_Ip_SetPreEnableClock (uint8 Instance, uint8 Channel, boolean Value)

  Set Prescaler Enable bit.
- void Emios_Pwm_Ip_SetBusSelected (uint8 Instance, uint8 Channel, Emios_Pwm_Ip_CounterBusSource↩ Type Value)

  Set Bus Select bits.
- void Emios_Pwm_Ip_SetClockPs (uint8 Instance, uint8 Channel, Emios_Pwm_Ip_InternalClkPsType Value)

  This function set the value of the prescaler on eMios channels.
- void Emios_Pwm_Ip_ComparatorTransferEnable (uint8 Instance, uint32 ChannelMask)

  The function shall Enable the output update for the corresponding Channel.
- void Emios_Pwm_Ip_ComparatorTransferDisable (uint8 Instance, uint32 ChannelMask)

  The function shall disable the output update for the corresponding Channel.
- void Emios_Pwm_Ip_SyncUpdate (uint8 Instance)

  This function updates the duty cycle and-or period for the specified PWM Channel. The value written does not take effect until calling SyncUpdate API.

7.1.2 Function Reference

7.1.2.1 Emios_Pwm_Ip_InitChannel()

```
void Emios_Pwm_Ip_InitChannel (
            uint8 Instance,
            Emios_Pwm_Ip_ChannelConfigType const * UserChCfg )
```

Initialize PWM Mode.

Parameters

in	*Instance*	The eMIOS group id
in	*UserChCfg*	A pointer to the PWM configuration structure

Returns

void

7.1.2.2 Emios_Pwm_Ip_DeInitChannel()

```
void Emios_Pwm_Ip_DeInitChannel (
            uint8 Instance,
            uint8 Channel )
```

Reset eMIOS Channel to GPIO mode (reset default)

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

void

7.1.2.3 Emios_Pwm_Ip_ForceMatchLeadingEdge()

```
void Emios_Pwm_Ip_ForceMatchLeadingEdge (
            uint8 Instance,
            uint8 Channel,
            boolean Enable )
```

Allow the software to force the output flip-flop to the level corresponding to a match on leading edge. The FLAG bit is not set.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Enable*	The Channel in Force Match Leading Edge or not

Returns

void

7.1.2.4 Emios_Pwm_Ip_ForceMatchTrailingEdge()

```
void Emios_Pwm_Ip_ForceMatchTrailingEdge (
            uint8 Instance,
            uint8 Channel,
            boolean Enable )
```

Allow the software to force the output flip-flop to the level corresponding to a match on trailing edge. The FLAG bit is not set.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Enable*	The Channel in Force Match Leading Edge or not

Returns

void

7.1.2.5 Emios_Pwm_Ip_GetPeriod()

```
uint16 Emios_Pwm_Ip_GetPeriod (
            uint8 Instance,
            uint8 Channel )
```

Get Period value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
out	*RetPeriod*	A pointer to return period value

Returns

uint16 Value of period

7.1.2.6 Emios_Pwm_Ip_SetPeriod()

```
void Emios_Pwm_Ip_SetPeriod (
            uint8 Instance,
            uint8 Channel,
            uint16 NewPeriod )
```

Set new Period value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*NewPeriod*	New Period value

Returns

void

7.1.2.7 Emios_Pwm_Ip_GetDutyCycle()

```
uint16 Emios_Pwm_Ip_GetDutyCycle (
            uint8 Instance,
            uint8 Channel )
```

Get Duty Cycle value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

uint16 Value of duty cycle

7.1.2.8 Emios_Pwm_Ip_SetDutyCycle()

```
Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_SetDutyCycle (
            uint8 Instance,
            uint8 Channel,
            uint16 NewDutyCycle )
```

Set new Duty Cycle value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*NewDutyCycle*	New duty cycle value

Returns

operation status

- EMIOS_PWM_IP_STATUS_SUCCESS : Operation was successful.
- EMIOS_PWM_IP_STATUS_ERROR : Operation failed, invalid input value.

7.1.2.9 Emios_Pwm_Ip_GetPhaseShift()

```
uint16 Emios_Pwm_Ip_GetPhaseShift (
            uint8 Instance,
            uint8 Channel )
```

Get Leading Edge Placement value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

uint16 Value of leading edge placement in counter bus time base

7.1.2.10 Emios_Pwm_Ip_SetPhaseShift()

```
Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_SetPhaseShift (
            uint8 Instance,
```

```
            uint8 Channel,
            uint16 PhaseShift )
```

Set new Leading edge placement value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*PhaseShift*	New Phase Shift value

Returns

void

7.1.2.11 Emios_Pwm_Ip_GetDeadTime()

```
uint16 Emios_Pwm_Ip_GetDeadTime (
            uint8 Instance,
            uint8 Channel )
```

Get dead time value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

uint16 Value of Dead Time

7.1.2.12 Emios_Pwm_Ip_SetDeadTime()

```
void Emios_Pwm_Ip_SetDeadTime (
            uint8 Instance,
            uint8 Channel,
            uint16 NewDeadTime )
```

Set new dead time value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*NewDeadTime*	New Dead Time value

Returns

void

7.1.2.13 Emios_Pwm_Ip_GetTriggerPlacement()

```
uint32 Emios_Pwm_Ip_GetTriggerPlacement (
            uint8 Instance,
            uint8 Channel )
```

Get Trigger Placement value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

uint32 Value of Trigger Placement

7.1.2.14 Emios_Pwm_Ip_SetTriggerPlacement()

```
void Emios_Pwm_Ip_SetTriggerPlacement (
            uint8 Instance,
            uint8 Channel,
            uint32 NewTriggerPlacement )
```

Set new Trigger Placement value in PWM mode.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*NewTriggerPlacement*	New Trigger Placement value

Returns

void

7.1.2.15 Emios_Pwm_Ip_ChannelEnterDebugMode()

```
Emios_Pwm_Ip_StatusType Emios_Pwm_Ip_ChannelEnterDebugMode (
            uint8 Instance,
            uint8 Channel )
```

Set a Channel enters freeze state, should be setting EMIOS_AllowEnterDebugMode first.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

operation status

- EMIOS_PWM_IP_STATUS_SUCCESS : Operation was successful.
- EMIOS_PWM_IP_STATUS_ERROR : Operation failed, invalid input value.
- EMIOS_PWM_IP_STATUS_ENABLE_GLOBAL_FRZ : Need call EMIOS_AllowEnterDebugMode first.

7.1.2.16 Emios_Pwm_Ip_ChannelStopDebugMode()

```
void Emios_Pwm_Ip_ChannelStopDebugMode (
            uint8 Instance,
            uint8 Channel )
```

Release a Channel from freeze state.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

void

7.1.2.17 Emios_Pwm_Ip_GetFlagRequest()

```
Emios_Pwm_Ip_InterruptType Emios_Pwm_Ip_GetFlagRequest (
            uint8 Instance,
            uint8 Channel )
```

Get the Unified Channel FLAG event generated. Interrupt or DMA request.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

The FLAG event response type

7.1.2.18 Emios_Pwm_Ip_SetFlagRequest()

```
void Emios_Pwm_Ip_SetFlagRequest (
            uint8 Instance,
            uint8 Channel,
            Emios_Pwm_Ip_InterruptType Event )
```

Allow the Unified Channel FLAG bit to generate an interrupt signal or a DMA request signal.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Event*	The FLAG event response type

Returns

void

7.1.2.19 Emios_Pwm_Ip_GetOutputState()

```
Emios_Pwm_Ip_OutputStateType Emios_Pwm_Ip_GetOutputState (
            uint8 Instance,
            uint8 Channel )
```

Get the Unified Channel output pin logic level.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

The Emios Channel output pin state HIGH/LOW

7.1.2.20 Emios_Pwm_Ip_SetOutputState()

```
void Emios_Pwm_Ip_SetOutputState (
            uint8 Instance,
            uint8 Channel,
            Emios_Pwm_Ip_OutputStateType OutputState )
```

Set the state of output pin.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*OutputState*	The state of output pin

Returns

void

7.1.2.21 Emios_Pwm_Ip_SetOutputToNormal()

```
void Emios_Pwm_Ip_SetOutputToNormal (
            uint8 Instance,
            uint8 Channel,
```

```
            uint16 DutyPercent,
            Emios_Pwm_Ip_PolarityType Polarity,
            Emios_Pwm_Ip_PwmModeType Mode )
```

Set the polarity and mode for current Channel as normal.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*DutyPercent*	The range of duty cycle value :0x00(0%) ->0x8000(100%)
in	*Polarity*	The polarity of Channel
in	*Mode*	Mode of Channel

Returns

void

7.1.2.22 Emios_Pwm_Ip_GetChannelMode()

```
Emios_Pwm_Ip_PwmModeType Emios_Pwm_Ip_GetChannelMode (
            uint8 Instance,
            uint8 Channel )
```

Get mode of operation of the Unified Channel.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

Emios_Pwm_Ip_PwmModeType

7.1.2.23 Emios_Pwm_Ip_GetMasterBusChannel()

```
uint8 Emios_Pwm_Ip_GetMasterBusChannel (
            uint8 Instance,
            uint8 Channel )
```

Get master bus Channel.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group

Returns

Emios_Pwm_Ip_PwmModeType

7.1.2.24 Emios_Pwm_Ip_SetPreEnableClock()

```
void Emios_Pwm_Ip_SetPreEnableClock (
            uint8 Instance,
            uint8 Channel,
            boolean Value )
```

Set Prescaler Enable bit.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Value*	The value to set • 0 Prescaler disabled (no clock) • 1 Prescaler enabled

Returns

void

7.1.2.25 Emios_Pwm_Ip_SetBusSelected()

```
void Emios_Pwm_Ip_SetBusSelected (
            uint8 Instance,
            uint8 Channel,
            Emios_Pwm_Ip_CounterBusSourceType Value )
```

Set Bus Select bits.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Value*	The value to set

Returns

void

7.1.2.26 Emios_Pwm_Ip_SetClockPs()

```
void Emios_Pwm_Ip_SetClockPs (
            uint8 Instance,
            uint8 Channel,
            Emios_Pwm_Ip_InternalClkPsType Value )
```

This function set the value of the prescaler on eMios channels.

Parameters

in	*Instance*	The eMIOS group id
in	*Channel*	The Channel in this eMIOS group
in	*Value*	The value to set

Returns

void

7.1.2.27 Emios_Pwm_Ip_ComparatorTransferEnable()

```
void Emios_Pwm_Ip_ComparatorTransferEnable (
            uint8 Instance,
            uint32 ChannelMask )
```

The function shall Enable the output update for the corresponding Channel.

Parameters

in	*Instance*	Instance of EMIOS used.
in	*ChannelMask*	EMIOS hardware mask Channel used.

7.1.2.28 Emios_Pwm_Ip_ComparatorTransferDisable()

```
void Emios_Pwm_Ip_ComparatorTransferDisable (
            uint8 Instance,
            uint32 ChannelMask )
```

The function shall disable the output update for the corresponding Channel.

Parameters

in	*Instance*	Instance of EMIOS used.
in	*ChannelMask*	EMIOS hardware mask Channel used.

7.1.2.29 Emios_Pwm_Ip_SyncUpdate()

```
void Emios_Pwm_Ip_SyncUpdate (
            uint8 Instance )
```

This function updates the duty cycle and-or period for the specified PWM Channel. The value written does not take effect until calling SyncUpdate API.

Parameters

Instance	eMIOS hardware module index

Returns

void

7.2 FlexIO Pwm IPL

7.2.1 Detailed Description

Function Reference

- Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_InitChannel (uint8 InstanceId, const Flexio_Pwm_Ip_ChannelConfigTyp *const UserCfg)

  Initialize a flexio channel in pwm mode.
- Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_DeInitChannel (uint8 InstanceId, uint8 Channel)

  Deinitialize a flexio channel.
- Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_UpdatePeriodDuty (uint8 InstanceId, uint8 Channel, uint16 Period, uint16 DutyCycle)

  Set a new value for duty cycle and period of the channel.
- boolean Flexio_Pwm_Ip_GetOutputState (uint8 InstanceId, uint8 Channel)

  Get the logic level of the channel ouput.
- Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_UpdateInterruptMode (uint8 InstanceId, uint8 Channel, Flexio_Pwm_Ip_InterruptType IrqMode)

  Update the interrupt mode for a channel.
- uint16 Flexio_Pwm_Ip_GetPeriod (uint8 InstanceId, uint8 Channel)

  Getting the period for a channel.

7.2.2 Function Reference

7.2.2.1 Flexio_Pwm_Ip_InitChannel()

```
Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_InitChannel (
            uint8 InstanceId,
            const Flexio_Pwm_Ip_ChannelConfigType *const UserCfg )
```

Initialize a flexio channel in pwm mode.

The function will initialize one timer and pin of the selected flexio channel in pwm mode, with the configuration of the user. The interrupts will be disabled.

Parameters

in	*InstanceId*	The Flexio instance id
in	*UserCfg*	The channel configuration for the selected Flexio instance

Returns

FLEXIO_PWM_IP_STATUS_SUCCESS - if the initialization was successfull

7.2.2.2 Flexio_Pwm_Ip_DeInitChannel()

```
Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_DeInitChannel (
            uint8 InstanceId,
            uint8 Channel )
```

Deinitialize a flexio channel.

The function will reset the timer and pin of the selected flexio channel.

Parameters

in	*InstanceId*	The Flexio instance id
in	*Channel*	The channel id for the selected Flexio instance

Returns

FLEXIO_PWM_IP_STATUS_SUCCESS - if the deinitialization was successfull

7.2.2.3 Flexio_Pwm_Ip_UpdatePeriodDuty()

```
Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_UpdatePeriodDuty (
            uint8 InstanceId,
            uint8 Channel,
            uint16 Period,
            uint16 DutyCycle )
```

Set a new value for duty cycle and period of the channel.

The function will update the selected flexio channel with the new values for the duty cycle and period.

Parameters

in	*InstanceId*	The Flexio instance id
in	*Channel*	The channel id for the selected Flexio instance
in	*Period*	The new value for the period
in	*DutyCycle*	The new value for the duty cycle

Returns

FLEXIO_PWM_IP_STATUS_SUCCESS - if the initialization was successfull

7.2.2.4 Flexio_Pwm_Ip_GetOutputState()

```
boolean Flexio_Pwm_Ip_GetOutputState (
            uint8 InstanceId,
            uint8 Channel )
```

Get the logic level of the channel ouput.

The function will return the logic level that the selected flexio channel is driving on on the output pin.

Parameters

in	*Instance↩ Id*	The Flexio instance id
in	*Channel*	The channel id for the selected Flexio instance

Returns

FLEXIO_PWM_IP_STATUS_SUCCESS - if the initialization was successfull

7.2.2.5 Flexio_Pwm_Ip_UpdateInterruptMode()

```
Flexio_Pwm_Ip_StatusType Flexio_Pwm_Ip_UpdateInterruptMode (
            uint8 InstanceId,
            uint8 Channel,
            Flexio_Pwm_Ip_InterruptType IrqMode )
```

Update the interrupt mode for a channel.

The function will set a new mode for the flag event response on the selected channel.

Parameters

in	*Instance↩ Id*	The Flexio instance id
in	*Channel*	The channel id for the selected Flexio instance
in	*IrqMode*	The new irq mode for the channel

Returns

FLEXIO_PWM_IP_STATUS_SUCCESS - if the initialization was successfull

7.2.2.6 Flexio_Pwm_Ip_GetPeriod()

```
uint16 Flexio_Pwm_Ip_GetPeriod (
            uint8 InstanceId,
            uint8 Channel )
```

Getting the period for a channel.

The function will get the period on the selected channel.

Parameters

in	*InstanceId*	The Flexio instance id
in	*Channel*	The channel id for the selected Flexio instance

Returns

uint16

7.3 Pwm Driver

7.3.1 Detailed Description

Data Structures

- struct Pwm_ChannelConfigType

  Pwm channel high level configuration structure. *More...*
- struct Pwm_ConfigType

  Pwm high level configuration structure. *More...*

Macros

- #define PWM_DUTY_CYCLE_100

  100% duty cycle
- #define PWM_E_INIT_FAILED

  API Pwm_Init service called with wrong parameter.
- #define PWM_E_UNINIT

  API service used without module initialization.
- #define PWM_E_PARAM_CHANNEL

  API service used with an invalid channel Identifier.
- #define PWM_E_PERIOD_UNCHANGEABLE

  Usage of unauthorized PWM service on PWM channel configured a fixed period.
- #define PWM_E_ALREADY_INITIALIZED

  API Pwm_Init service called while the PWM driver has already been initialized.
- #define PWM_E_PARAM_POINTER

  Generated when a NULL_PTR pointer is passed to Pwm_GetVersionInfo function.
- #define PWM_E_NOT_DISENGAGED

  Generated when Pwm_SetPowerState is called while the PWM module is still in use.
- #define PWM_E_POWER_STATE_NOT_SUPPORTED

  The requested power state is not supported by the PWM module.
- #define PWM_E_TRANSITION_NOT_POSSIBLE

  Generated The requested power state is not reachable from the current one.
- #define PWM_E_PERIPHERAL_NOT_PREPARED

  Generated when Pwm_SetPowerState has been called without having called the API Pwm_PreparePowerState before.
- #define PWM_E_PERIODVALUE

  Pwm_SetPeriodAndDuty called with invalid period range.
- #define PWM_E_PARAM_NOTIFICATION

  Invalid polarity selected for edge notification.
- #define PWM_E_PARAM_NOTIFICATION_NULL

  NULL_PTR function is configured as notification callback.
- #define PWM_E_DUTYCYCLE_RANGE

  Pwm_SetDutyCycle or Pwm_SetPeriodAndDuty called with invalid duty cycle range.
- #define PWM_E_COUNTERBUS

Generated when Pwm_SetCounterBus is called with an invalid Bus.

- #define PWM_E_CHANNEL_OFFSET_VALUE

  Generated when the configured offset for the OPWMB channel is more than the period of the associated MCB channel.
- #define PWM_E_OPWMB_CHANNEL_OFFSET_DUTYCYCLE_RANGE

  Generated when the requested offset value plus the current requested duty cycle leads to programming event B over the Period value leading to unexpected behavior of the PWM signal.
- #define PWM_E_PARAM_INSTANCE

  Generated when the module id is more than the number of module that supported by this platform.
- #define PWM_E_OPWMT_CHANNEL_TRIGGER_RANGE

  Generated when the configured trigger value for the OPWMT channel is equal or greater than the period of the channel.
- #define PWM_E_SET_CHANNEL_OUTPUT

  Generated when the output state value for the SetChannelOutput of the channel.
- #define PWM_E_UNEXPECTED_ISR

  Generated when an ISR has been triggered.
- #define PWM_E_PARAM_PHASESHIFT_RANGE

  Generated when requested phase shift value greater than 0x4000 (50%)
- #define PWM_E_CHANNEL_PHASE_SHIFT_NOT_SUPPORTED

  Generated when given channel does not support phase shift feature.
- #define PWM_E_DUTY_SYNCHRONOUS_NOT_SUPPORTED

  Generated when given channel does not support duty synchronous feature. (PHASE_SHIFTED_SYNCED and P↩ HASE_SHIFTED_COMPLEMENTARY)
- #define PWM_E_TRIGGER_MASK

  Generated when bit mask is not compatible with hardware register.
- #define PWM_E_FORCE_OUTPUT_NOT_SUPPORTED

  Generated when given channel does not support force output to zero feature.
- #define PWM_E_FORCE_OUT

  Generated when given channel does not support set force out feature.
- #define PWM_E_PARAM_CONFIG

  Generated when the requested resource is configured to be unavailable on the current core.
- #define PWM_E_DEADTIME_RANGE

  Generated when the configured dead time value is not valid.
- #define PWM_E_SETOUTPUTTOIDLE_NOT_SUPPORTED

  Generated when the configured dead time value is not valid.
- #define PWM_INIT_ID

  API service ID of Pwm_Init function.
- #define PWM_DEINIT_ID

  API service ID of Pwm_DeInit function.
- #define PWM_SETDUTYCYCLE_ID

  API service ID of Pwm_SetDutyCycle function.
- #define PWM_SETPERIODANDDUTY_ID

  API service ID of Pwm_SetPeriodAndDuty function.
- #define PWM_SETOUTPUTTOIDLE_ID

  API service ID of Pwm_SetOutputToIdle function.
- #define PWM_GETOUTPUTSTATE_ID

  API service ID of Pwm_GetOutputState function.
- #define PWM_DISABLENOTIFICATION_ID

API service ID of Pwm_DisableNotification function.

- #define PWM_ENABLENOTIFICATION_ID

  API service ID of Pwm_EnableNotification function.
- #define PWM_GETVERSIONINFO_ID

  API service ID of Pwm_GetVersionInfo function.
- #define PWM_SETPOWERSTATE_ID

  API service ID of Pwm_SetPowerState function.
- #define PWM_GETCURRENTPOWERSTATE_ID

  API service ID of Pwm_GetCurrentPowerState function.
- #define PWM_GETTARGETPOWERSTATE_ID

  API service ID of Pwm_GetTargetPowerState function.
- #define PWM_PREPAREPOWERSTATE_ID

  API service ID of Pwm_PreparePowerState function.
- #define PWM_MAIN_POWERTRANSITIONMANAGER_ID

  API service ID of Pwm_Main_PowerTransitionManager function.
- #define PWM_GETCHANNELSTATE_ID

  API service ID of Pwm_GetChannelState function.
- #define PWM_FORCEOUTPUTTOZERO_ID

  API service ID of Pwm_ForceOutputToZero function.
- #define PWM_SETCOUNTERBUS_ID

  API service ID of Pwm_SetCounterBus function.
- #define PWM_SETCHANNELOUTPUT_ID

  API service ID of Pwm_SetChannelOutput function.
- #define PWM_SETTRIGGERDELAY_ID

  API service ID of Pwm_SetTriggerDelay function.
- #define PWM_SETCLOCKMODE_ID

  API service ID of Pwm_SetClockMode function.
- #define PWM_SYNCUPDATE_ID

  API service ID of Pwm_SyncUpdate function.
- #define PWM_SETPERIODANDDUTY_NO_UPDATE_ID

  API service ID of Pwm_SetPeriodAndDuty_NoUpdate function.
- #define PWM_SETDUTYCYCLE_NO_UPDATE_ID

  API service ID of Pwm_SetDutyCycle_NoUpdate function.
- #define PWM_SETCHANNELDEADTIME_ID

  API service ID of Pwm_SetChannelDeadTime function.
- #define PWM_SETPHASESHIFT_ID

  API service ID of Pwm_SetPhaseShift function.
- #define PWM_SETPHASESHIFTNOUPDATE_ID

  API service ID of Pwm_SetPhaseShift function.
- #define PWM_ENABLETRIGGER_ID

  API service ID of Pwm_EnableTrigger function.
- #define PWM_DISABLETRIGGER_ID

  API service ID of Pwm_DisableTrigger function.
- #define PWM_RESETCOUNTERENABLE_ID

  API service ID of Pwm_ResetCounterEnable function.
- #define PWM_RESETCOUNTERDISABLE_ID

  API service ID of Pwm_ResetCounterDisable function.

- #define PWM_MASKOUTPUT_ID
  API service ID of Pwm_MaskOutputs function.
- #define PWM_UNMASKOUTPUT_ID
  API service ID of Pwm_UnMaskOutputs function.
- #define PWM_DISABLERELOADNOTIF_ID
  API service ID of Pwm_DisableReloadNotification function.
- #define PWM_ENABLERELOADNOTIF_ID
  API service ID of Pwm_EnableReloadNotification function.
- #define PWM_SETCHANNELFORCEOUT_ID
  API service ID of Pwm_SetChannelForceOut function.
- #define PWM_SETDUTYPHASESHIFT_ID
  API service ID of Pwm_SetDutyPhaseShift function.

Types Reference

- typedef uint8 Pwm_ChannelType
  PWM channel type.
- typedef uint8 Pwm_InstanceType
  PWM channel type.
- typedef Pwm_Ipw_PeriodType Pwm_PeriodType
  PWM period type.

Enum Reference

- enum Pwm_OutputStateType
  Output signal level.
- enum Pwm_EdgeNotificationType
  Edge notification type.
- enum Pwm_ChannelClassType
  PWM channel class type.
- enum Pwm_PowerStateType
  Power state type.
- enum Pwm_PowerStateRequestResultType
  Result of power state type.
- enum Pwm_PrescalerType
  Prescaler type.

Function Reference

- void Pwm_Init (const Pwm_ConfigType *ConfigPtr)
  This function initializes the Pwm driver.

7.3.2 Data Structure Documentation

7.3.2.1 struct Pwm_ChannelConfigType

Pwm channel high level configuration structure.

Definition at line 720 of file Pwm.h.

Data Fields

- const Pwm_ChannelType ChannelId
  Id for the logical channel.
- const Pwm_ChannelClassType PwmChannelClass
  Channel class type: Variable/Fixed period.
- const Pwm_IpwChannelConfigType IpwChannelCfg
  The type of ip channel configured.
- const Pwm_OutputStateType ChannelIdleState
  The state of the channel output in idle mode.

7.3.2.1.1 Field Documentation

7.3.2.1.1.1 ChannelId `const Pwm_ChannelType ChannelId`

Id for the logical channel.

Definition at line 723 of file Pwm.h.

7.3.2.1.1.2 PwmChannelClass `const Pwm_ChannelClassType PwmChannelClass`

Channel class type: Variable/Fixed period.

Definition at line 725 of file Pwm.h.

7.3.2.1.1.3 IpwChannelCfg `const Pwm_IpwChannelConfigType IpwChannelCfg`

The type of ip channel configured.

Definition at line 727 of file Pwm.h.

7.3.2.1.1.4 ChannelIdleState `const Pwm_OutputStateType ChannelIdleState`

The state of the channel output in idle mode.

Definition at line 729 of file Pwm.h.

7.3.2.2 struct Pwm_ConfigType

Pwm high level configuration structure.

Definition at line 739 of file Pwm.h.

Data Fields

- const Pwm_ChannelType NumChannels
  Number of Pwm configured channels.
- const Pwm_ChannelConfigType(* PwmChannelsConfig)[]
  Pointer to the list of Pwm configured channels.

7.3.2.2.1 Field Documentation

7.3.2.2.1.1 NumChannels `const Pwm_ChannelType NumChannels`

Number of Pwm configured channels.

Definition at line 746 of file Pwm.h.

7.3.2.2.1.2 PwmChannelsConfig `const Pwm_ChannelConfigType(* PwmChannelsConfig)[]`

Pointer to the list of Pwm configured channels.

Definition at line 748 of file Pwm.h.

7.3.3 Macro Definition Documentation

7.3.3.1 PWM_DUTY_CYCLE_100

```
#define PWM_DUTY_CYCLE_100
```

100% duty cycle

Errors and exceptions that will be detected by the PWM driver generated when Pwm_SetDutyCycle or Pwm_↩ SetPeriodAndDuty are called with a value for duty cycle out of valid range [0x0000, 0x8000]

Definition at line 145 of file Pwm.h.

7.3.3.2 PWM_E_INIT_FAILED

```
#define PWM_E_INIT_FAILED
```

API Pwm_Init service called with wrong parameter.

Errors and exceptions that will be detected by the PWM driver

Definition at line 152 of file Pwm.h.

7.3.3.3 PWM_E_UNINIT

```
#define PWM_E_UNINIT
```

API service used without module initialization.

Errors and exceptions that will be detected by the PWM driver

Definition at line 158 of file Pwm.h.

7.3.3.4 PWM_E_PARAM_CHANNEL

```
#define PWM_E_PARAM_CHANNEL
```

API service used with an invalid channel Identifier.

Errors and exceptions that will be detected by the PWM driver

Definition at line 164 of file Pwm.h.

7.3.3.5 PWM_E_PERIOD_UNCHANGEABLE

```
#define PWM_E_PERIOD_UNCHANGEABLE
```

Usage of unauthorized PWM service on PWM channel configured a fixed period.

Errors and exceptions that will be detected by the PWM driver

Definition at line 170 of file Pwm.h.

7.3.3.6 PWM_E_ALREADY_INITIALIZED

```
#define PWM_E_ALREADY_INITIALIZED
```

API Pwm_Init service called while the PWM driver has already been initialized.

Errors and exceptions that will be detected by the PWM driver

Definition at line 176 of file Pwm.h.

7.3.3.7 PWM_E_PARAM_POINTER

```
#define PWM_E_PARAM_POINTER
```

Generated when a NULL_PTR pointer is passed to Pwm_GetVersionInfo function.

Errors and exceptions that will be detected by the PWM driver

Definition at line 182 of file Pwm.h.

7.3.3.8 PWM_E_NOT_DISENGAGED

```
#define PWM_E_NOT_DISENGAGED
```

Generated when Pwm_SetPowerState is called while the PWM module is still in use.

Errors and exceptions that will be detected by the PWM driver

Definition at line 188 of file Pwm.h.

7.3.3.9 PWM_E_POWER_STATE_NOT_SUPPORTED

```
#define PWM_E_POWER_STATE_NOT_SUPPORTED
```

The requested power state is not supported by the PWM module.

Errors and exceptions that will be detected by the PWM driver

Definition at line 194 of file Pwm.h.

7.3.3.10 PWM_E_TRANSITION_NOT_POSSIBLE

```
#define PWM_E_TRANSITION_NOT_POSSIBLE
```

Generated The requested power state is not reachable from the current one.

Errors and exceptions that will be detected by the PWM driver

Definition at line 200 of file Pwm.h.

7.3.3.11 PWM_E_PERIPHERAL_NOT_PREPARED

```
#define PWM_E_PERIPHERAL_NOT_PREPARED
```

Generated when Pwm_SetPowerState has been called without having called the API Pwm_PreparePowerState before.

Errors and exceptions that will be detected by the PWM driver

Definition at line 207 of file Pwm.h.

7.3.3.12 PWM_E_PERIODVALUE

```
#define PWM_E_PERIODVALUE
```

Pwm_SetPeriodAndDuty called with invalid period range.

Generated when Pwm_SetPeriodAndDuty is called with a value for period out of valid range [0x0000, PWM_M↩AX_PERIOD]

Definition at line 214 of file Pwm.h.

7.3.3.13 PWM_E_PARAM_NOTIFICATION

```
#define PWM_E_PARAM_NOTIFICATION
```

Invalid polarity selected for edge notification.

Will be generated when an invalid polarity, edge notification is requested for one PWM channel. Due to the limitations that are present in the eMIOS implementation not all the polarity notifications combinations can be supported.

Definition at line 222 of file Pwm.h.

7.3.3.14 PWM_E_PARAM_NOTIFICATION_NULL

```
#define PWM_E_PARAM_NOTIFICATION_NULL
```

NULL_PTR function is configured as notification callback.

Will be generated when a NULL_PTR function is configured as notification callback for one PWM channel and Pwm_EnableNotification is called for that channel

Definition at line 229 of file Pwm.h.

7.3.3.15 PWM_E_DUTYCYCLE_RANGE

```
#define PWM_E_DUTYCYCLE_RANGE
```

Pwm_SetDutyCycle or Pwm_SetPeriodAndDuty called with invalid duty cycle range.

Generated when Pwm_SetDutyCycle or Pwm_SetPeriodAndDuty are called with a value for duty cycle out of valid range [0x0000, 0x8000]

Definition at line 236 of file Pwm.h.

7.3.3.16 PWM_E_COUNTERBUS

```
#define PWM_E_COUNTERBUS
```

Generated when Pwm_SetCounterBus is called with an invalid Bus.

Errors and exceptions that will be detected by the PWM driver

Definition at line 242 of file Pwm.h.

7.3.3.17 PWM_E_CHANNEL_OFFSET_VALUE

```
#define PWM_E_CHANNEL_OFFSET_VALUE
```

Generated when the configured offset for the OPWMB channel is more than the period of the associated MCB channel.

Errors and exceptions that will be detected by the PWM driver

Definition at line 249 of file Pwm.h.

7.3.3.18 PWM_E_OPWMB_CHANNEL_OFFSET_DUTYCYCLE_RANGE

```
#define PWM_E_OPWMB_CHANNEL_OFFSET_DUTYCYCLE_RANGE
```

Generated when the requested offset value plus the current requested duty cycle leads to programming event B over the Period value leading to unexpected behavior of the PWM signal.

Errors and exceptions that will be detected by the PWM driver

Definition at line 256 of file Pwm.h.

7.3.3.19 PWM_E_PARAM_INSTANCE

```
#define PWM_E_PARAM_INSTANCE
```

Generated when the module id is more than the number of module that supported by this platform.

Errors and exceptions that will be detected by the PWM driver

Definition at line 263 of file Pwm.h.

7.3.3.20 PWM_E_OPWMT_CHANNEL_TRIGGER_RANGE

```
#define PWM_E_OPWMT_CHANNEL_TRIGGER_RANGE
```

Generated when the configured trigger value for the OPWMT channel is equal or greater than the period of the channel.

Errors and exceptions that will be detected by the PWM driver

Definition at line 270 of file Pwm.h.

7.3.3.21 PWM_E_SET_CHANNEL_OUTPUT

```
#define PWM_E_SET_CHANNEL_OUTPUT
```

Generated when the output state value for the SetChannelOutput of the channel.

Errors and exceptions that will be detected by the PWM driver

Definition at line 277 of file Pwm.h.

7.3.3.22 PWM_E_UNEXPECTED_ISR

```
#define PWM_E_UNEXPECTED_ISR
```

Generated when an ISR has been triggered.

1. when the driver is not initialized
2. for a HW channel that is not used by any logic channel
3. for a logic channel that has no notification configured

Errors and exceptions that will be detected by the PWM driver

Definition at line 286 of file Pwm.h.

7.3.3.23 PWM_E_PARAM_PHASESHIFT_RANGE

```
#define PWM_E_PARAM_PHASESHIFT_RANGE
```

Generated when requested phase shift value greater than 0x4000 (50%)

Pwm_SetPhaseShift only works with Combine channel (COMBINED_SYNCED or COMBINED_COMPLEME↩ NTARY) Which do not support matching at next cycle. The duty cycle is always fixed value at 50%, so Phase Shift cannot greater than 50%

Definition at line 294 of file Pwm.h.

7.3.3.24 PWM_E_CHANNEL_PHASE_SHIFT_NOT_SUPPORTED

```
#define PWM_E_CHANNEL_PHASE_SHIFT_NOT_SUPPORTED
```

Generated when given channel does not support phase shift feature.

For FTM, only combine mode (COMBINED_SYNCED and COMBINED_COMPLEMENTARY) is supported. For eMIOS, only OPWMB, OPWM and OPWMT mode is supported.

Definition at line 301 of file Pwm.h.

7.3.3.25 PWM_E_DUTY_SYNCHRONOUS_NOT_SUPPORTED

```
#define PWM_E_DUTY_SYNCHRONOUS_NOT_SUPPORTED
```

Generated when given channel does not support duty synchronous feature. (PHASE_SHIFTED_SYNCED and PHASE_SHIFTED_COMPLEMENTARY)

For FTM, please note that Modified Combine mode does not support synchronous update Cn Therefore Pwm_↩ SetDutyCycle_NoUpdate and Pwm_SetPeriodAndDuty_NoUpdate should not be called in this case. For eMIOS, please note that channels using DAOC mode or channels in idle state does not support.

Definition at line 312 of file Pwm.h.

7.3.3.26 PWM_E_TRIGGER_MASK

```
#define PWM_E_TRIGGER_MASK
```

Generated when bit mask is not compatible with hardware register.

Definition at line 318 of file Pwm.h.

7.3.3.27 PWM_E_FORCE_OUTPUT_NOT_SUPPORTED

```
#define PWM_E_FORCE_OUTPUT_NOT_SUPPORTED
```

Generated when given channel does not support force output to zero feature.

Only channels in FlexPWM and FTM module are supported.

Definition at line 324 of file Pwm.h.

7.3.3.28 PWM_E_FORCE_OUT

```
#define PWM_E_FORCE_OUT
```

Generated when given channel does not support set force out feature.

Only channels in FlexPWM is supported.

Definition at line 330 of file Pwm.h.

7.3.3.29 PWM_E_PARAM_CONFIG

```
#define PWM_E_PARAM_CONFIG
```

Generated when the requested resource is configured to be unavailable on the current core.

Only multi-core configuration is available.

Definition at line 336 of file Pwm.h.

7.3.3.30 PWM_E_DEADTIME_RANGE

```
#define PWM_E_DEADTIME_RANGE
```

Generated when the configured dead time value is not valid.

Errors and exceptions that will be detected by the PWM driver

Definition at line 342 of file Pwm.h.

7.3.3.31 PWM_E_SETOUTPUTTOIDLE_NOT_SUPPORTED

```
#define PWM_E_SETOUTPUTTOIDLE_NOT_SUPPORTED
```

Generated when the configured dead time value is not valid.

Errors and exceptions that will be detected by the PWM driver

Definition at line 348 of file Pwm.h.

7.3.3.32 PWM_INIT_ID

```
#define PWM_INIT_ID
```

API service ID of Pwm_Init function.

Parameters used when raising an error/exception

Definition at line 354 of file Pwm.h.

7.3.3.33 PWM_DEINIT_ID

```
#define PWM_DEINIT_ID
```

API service ID of Pwm_DeInit function.

Parameters used when raising an error/exception

Definition at line 360 of file Pwm.h.

7.3.3.34 PWM_SETDUTYCYCLE_ID

```
#define PWM_SETDUTYCYCLE_ID
```

API service ID of Pwm_SetDutyCycle function.

Parameters used when raising an error/exception

Definition at line 366 of file Pwm.h.

7.3.3.35 PWM_SETPERIODANDDUTY_ID

```
#define PWM_SETPERIODANDDUTY_ID
```

API service ID of Pwm_SetPeriodAndDuty function.

Parameters used when raising an error/exception

Definition at line 372 of file Pwm.h.

7.3.3.36 PWM_SETOUTPUTTOIDLE_ID

```
#define PWM_SETOUTPUTTOIDLE_ID
```

API service ID of Pwm_SetOutputToIdle function.

Parameters used when raising an error/exception

Definition at line 378 of file Pwm.h.

7.3.3.37 PWM_GETOUTPUTSTATE_ID

```
#define PWM_GETOUTPUTSTATE_ID
```

API service ID of Pwm_GetOutputState function.

Parameters used when raising an error/exception

Definition at line 384 of file Pwm.h.

7.3.3.38 PWM_DISABLENOTIFICATION_ID

```
#define PWM_DISABLENOTIFICATION_ID
```

API service ID of Pwm_DisableNotification function.

Parameters used when raising an error/exception

Definition at line 390 of file Pwm.h.

7.3.3.39 PWM_ENABLENOTIFICATION_ID

```
#define PWM_ENABLENOTIFICATION_ID
```

API service ID of Pwm_EnableNotification function.

Parameters used when raising an error/exception

Definition at line 396 of file Pwm.h.

7.3.3.40 PWM_GETVERSIONINFO_ID

```
#define PWM_GETVERSIONINFO_ID
```

API service ID of Pwm_GetVersionInfo function.

Parameters used when raising an error/exception

Definition at line 402 of file Pwm.h.

7.3.3.41 PWM_SETPOWERSTATE_ID

```
#define PWM_SETPOWERSTATE_ID
```

API service ID of Pwm_SetPowerState function.

Parameters used when raising an error/exception

Definition at line 408 of file Pwm.h.

7.3.3.42 PWM_GETCURRENTPOWERSTATE_ID

```
#define PWM_GETCURRENTPOWERSTATE_ID
```

API service ID of Pwm_GetCurrentPowerState function.

Parameters used when raising an error/exception

Definition at line 414 of file Pwm.h.

7.3.3.43 PWM_GETTARGETPOWERSTATE_ID

```
#define PWM_GETTARGETPOWERSTATE_ID
```

API service ID of Pwm_GetTargetPowerState function.

Parameters used when raising an error/exception

Definition at line 420 of file Pwm.h.

7.3.3.44 PWM_PREPAREPOWERSTATE_ID

```
#define PWM_PREPAREPOWERSTATE_ID
```

API service ID of Pwm_PreparePowerState function.

Parameters used when raising an error/exception

Definition at line 426 of file Pwm.h.

7.3.3.45 PWM_MAIN_POWERTRANSITIONMANAGER_ID

```
#define PWM_MAIN_POWERTRANSITIONMANAGER_ID
```

API service ID of Pwm_Main_PowerTransitionManager function.

Parameters used when raising an error/exception

Definition at line 432 of file Pwm.h.

7.3.3.46 PWM_GETCHANNELSTATE_ID

```
#define PWM_GETCHANNELSTATE_ID
```

API service ID of Pwm_GetChannelState function.

Parameters used when raising an error/exception

Definition at line 438 of file Pwm.h.

7.3.3.47 PWM_FORCEOUTPUTTOZERO_ID

```
#define PWM_FORCEOUTPUTTOZERO_ID
```

API service ID of Pwm_ForceOutputToZero function.

Parameters used when raising an error/exception

Definition at line 444 of file Pwm.h.

7.3.3.48 PWM_SETCOUNTERBUS_ID

```
#define PWM_SETCOUNTERBUS_ID
```

API service ID of Pwm_SetCounterBus function.

Parameters used when raising an error/exception

Definition at line 450 of file Pwm.h.

7.3.3.49 PWM_SETCHANNELOUTPUT_ID

```
#define PWM_SETCHANNELOUTPUT_ID
```

API service ID of Pwm_SetChannelOutput function.

Parameters used when raising an error/exception

Definition at line 456 of file Pwm.h.

7.3.3.50 PWM_SETTRIGGERDELAY_ID

```
#define PWM_SETTRIGGERDELAY_ID
```

API service ID of Pwm_SetTriggerDelay function.

Parameters used when raising an error/exception

Definition at line 462 of file Pwm.h.

7.3.3.51 PWM_SETCLOCKMODE_ID

```
#define PWM_SETCLOCKMODE_ID
```

API service ID of Pwm_SetClockMode function.

Parameters used when raising an error/exception

Definition at line 468 of file Pwm.h.

7.3.3.52 PWM_SYNCUPDATE_ID

```
#define PWM_SYNCUPDATE_ID
```

API service ID of Pwm_SyncUpdate function.

Parameters used when raising an error/exception

Definition at line 474 of file Pwm.h.

7.3.3.53 PWM_SETPERIODANDDUTY_NO_UPDATE_ID

```
#define PWM_SETPERIODANDDUTY_NO_UPDATE_ID
```

API service ID of Pwm_SetPeriodAndDuty_NoUpdate function.

Parameters used when raising an error/exception

Definition at line 480 of file Pwm.h.

7.3.3.54 PWM_SETDUTYCYCLE_NO_UPDATE_ID

```
#define PWM_SETDUTYCYCLE_NO_UPDATE_ID
```

API service ID of Pwm_SetDutyCycle_NoUpdate function.

Parameters used when raising an error/exception

Definition at line 486 of file Pwm.h.

7.3.3.55 PWM_SETCHANNELDEADTIME_ID

```
#define PWM_SETCHANNELDEADTIME_ID
```

API service ID of Pwm_SetChannelDeadTime function.

Parameters used when raising an error/exception

Definition at line 492 of file Pwm.h.

7.3.3.56 PWM_SETPHASESHIFT_ID

```
#define PWM_SETPHASESHIFT_ID
```

API service ID of Pwm_SetPhaseShift function.

Parameters used when raising an error/exception

Definition at line 498 of file Pwm.h.

7.3.3.57 PWM_SETPHASESHIFTNOUPDATE_ID

```
#define PWM_SETPHASESHIFTNOUPDATE_ID
```

API service ID of Pwm_SetPhaseShift function.

Parameters used when raising an error/exception

Definition at line 504 of file Pwm.h.

7.3.3.58 PWM_ENABLETRIGGER_ID

```
#define PWM_ENABLETRIGGER_ID
```

API service ID of Pwm_EnableTrigger function.

Parameters used when raising an error/exception

Definition at line 510 of file Pwm.h.

7.3.3.59 PWM_DISABLETRIGGER_ID

```
#define PWM_DISABLETRIGGER_ID
```

API service ID of Pwm_DisableTrigger function.

Parameters used when raising an error/exception

Definition at line 516 of file Pwm.h.

7.3.3.60 PWM_RESETCOUNTERENABLE_ID

```
#define PWM_RESETCOUNTERENABLE_ID
```

API service ID of Pwm_ResetCounterEnable function.

Parameters used when raising an error/exception

Definition at line 522 of file Pwm.h.

7.3.3.61 PWM_RESETCOUNTERDISABLE_ID

```
#define PWM_RESETCOUNTERDISABLE_ID
```

API service ID of Pwm_ResetCounterDisable function.

Parameters used when raising an error/exception

Definition at line 528 of file Pwm.h.

7.3.3.62 PWM_MASKOUTPUT_ID

```
#define PWM_MASKOUTPUT_ID
```

API service ID of Pwm_MaskOutputs function.

Parameters used when raising an error/exception

Definition at line 534 of file Pwm.h.

7.3.3.63 PWM_UNMASKOUTPUT_ID

```
#define PWM_UNMASKOUTPUT_ID
```

API service ID of Pwm_UnMaskOutputs function.

Parameters used when raising an error/exception

Definition at line 540 of file Pwm.h.

7.3.3.64 PWM_DISABLERELOADNOTIF_ID

```
#define PWM_DISABLERELOADNOTIF_ID
```

API service ID of Pwm_DisableReloadNotification function.

Parameters used when raising an error/exception

Definition at line 546 of file Pwm.h.

7.3.3.65 PWM_ENABLERELOADNOTIF_ID

```
#define PWM_ENABLERELOADNOTIF_ID
```

API service ID of Pwm_EnableReloadNotification function.

Parameters used when raising an error/exception

Definition at line 552 of file Pwm.h.

7.3.3.66 PWM_SETCHANNELFORCEOUT_ID

```
#define PWM_SETCHANNELFORCEOUT_ID
```

API service ID of Pwm_SetChannelForceOut function.

Parameters used when raising an error/exception

Definition at line 558 of file Pwm.h.

7.3.3.67 PWM_SETDUTYPHASESHIFT_ID

```
#define PWM_SETDUTYPHASESHIFT_ID
```

API service ID of Pwm_SetDutyPhaseShift function.

Parameters used when raising an error/exception

Definition at line 564 of file Pwm.h.

7.3.4 Types Reference

7.3.4.1 Pwm_ChannelType

```
typedef uint8 Pwm_ChannelType
```

PWM channel type.

Definition at line 694 of file Pwm.h.

7.3.4.2 Pwm_InstanceType

```
typedef uint8 Pwm_InstanceType
```

PWM channel type.

Definition at line 700 of file Pwm.h.

7.3.4.3 Pwm_PeriodType

```
typedef Pwm_Ipw_PeriodType Pwm_PeriodType
```

PWM period type.

Definition at line 706 of file Pwm.h.

7.3.5 Enum Reference

7.3.5.1 Pwm_OutputStateType

```
enum Pwm_OutputStateType
```

Output signal level.

This enumeration specifies the return type of Pwm_GetOutputState

Enumerator

PWM_LOW	PWM level is logic low.
PWM_HIGH	PWM level is logic high.

Definition at line 575 of file Pwm.h.

7.3.5.2 Pwm_EdgeNotificationType

```
enum Pwm_EdgeNotificationType
```

Edge notification type.

This enumeration defines the type of edge transition that can generate a notification

Enumerator

PWM_FALLING_EDGE	A notification will be generated on the falling edge.
PWM_RISING_EDGE	A notification will be generated on the rising edge.
PWM_BOTH_EDGES	A notification will be generated on any state transition.

Definition at line 589 of file Pwm.h.

7.3.5.3 Pwm_ChannelClassType

`enum Pwm_ChannelClassType`

PWM channel class type.

This field will specify what parameters can be altered for the selected channel

Enumerator

PWM_VARIABLE_PERIOD	The period and duty cycle can be altered.
PWM_FIXED_PERIOD	Only the duty cycle can be altered.
PWM_FIXED_PERIOD_SHIFTED	Only the duty cycle can be altered.

Definition at line 605 of file Pwm.h.

7.3.5.4 Pwm_PowerStateType

`enum Pwm_PowerStateType`

Power state type.

Power state currently active or set as target power state.

Enumerator

PWM_FULL_POWER	PWM full power mode.
PWM_LOW_POWER	PWM low power mode.
PWM_NODEFINE_POWER	PWM no define power mode.

Definition at line 621 of file Pwm.h.

7.3.5.5 Pwm_PowerStateRequestResultType

```
enum Pwm_PowerStateRequestResultType
```

Result of power state type.

Result of the requests related to power state transitions.

Enumerator

PWM_SERVICE_ACCEPTED	Power state change executed.
PWM_NOT_INIT	Module not initialized.
PWM_SEQUENCE_ERROR	Wrong API call sequence.
PWM_HW_FAILURE	The HW module has a failure which prevents it to enter the required power state.
PWM_POWER_STATE_NOT_SUPP	Module does not support the requested power state.
PWM_TRANS_NOT_POSSIBLE	Module cannot transition directly from the current power state to the requested power state.

Definition at line 637 of file Pwm.h.

7.3.5.6 Pwm_PrescalerType

```
enum Pwm_PrescalerType
```

Prescaler type.

This enumeration specifies the possible types of prescalers used to configure base-clock timers

Enumerator

PWM_PRIMARY_PRESCALER	Selected value is the default/primary prescaler.
PWM_ALTERNATIVE_PRESCALER	Selected value is the alternative configured prescaler.

Definition at line 658 of file Pwm.h.

7.3.6 Function Reference

7.3.6.1 Pwm_Init()

```
void Pwm_Init (
            const Pwm_ConfigType * ConfigPtr )
```

This function initializes the Pwm driver.

The function Pwm_Init shall initialize all internals variables and the used PWM structure of the microcontroller according to the parameters specified in configPtr. If the duty cycle parameter equals:

```
- 0% or 100% : Then the PWM output signal shall be in the state according
    to the configured polarity parameter;

- >0% and <100%: Then the PWM output signal shall be modulated according
    to parameters period, duty cycle and configured polarity.
```

The function Pwm_SetDutyCycle shall update the duty cycle always at the end of the period if supported by the implementation and configured with PwmDutycycleUpdatedEndperiod.

The driver shall avoid spikes on the PWM output signal when updating the PWM period and duty.

If development error detection for the Pwm module is enabled, the PWM functions shall check the channel class type and raise development error PWM_E_PERIOD_UNCHANGEABLE if the PWM channel is not declared as a variable period type.

If development error detection for the Pwm module is enabled, the PWM functions shall check the parameter channelNumber and raise development error PWM_E_PARAM_CHANNEL if the parameter channelNumber is invalid.

If development error detection for the Pwm module is enabled, when a development error occurs, the corresponding PWM function shall:

```
- Report the error to the Development Error Tracer.
- Skip the desired functionality in order to avoid any corruptions of
    data or hardware registers (this means leave the function without any actions).
- Return pwm level low for the function Pwm_GetOutputState.
```

The function Pwm_Init shall disable all notifications. The reason is that the users of these notifications may not be ready. They can call Pwm_EnableNotification to start notifications.

The function Pwm_Init shall only initialize the configured resources and shall not touch resources that are not configured in the configuration file.

If the PwmDevErorDetect switch is enabled, API parameter checking is enabled. The detailed description of the detected errors can be found in chapter Error classification and chapter API specification (see PWM_SWS).

If development error detection is enabled, calling the routine Pwm_Init while the PWM driver and hardware are already initialized will cause a development error PWM_E_ALREADY_INITIALIZED. The desired functionality shall be left without any action.

For pre-compile and link time configuration variants, a NULL pointer shall be passed to the initialization routine. In this case the check for this NULL pointer has to be omitted.

If development error detection for the Pwm module is enabled, if any function (except Pwm_Init) is called before Pwm_Init has been called, the called function shall raise development error PWM_E_UNINIT.

Parameters

in	*ConfigPtr*	Pointer to PWM top configuration structure

Returns

void

Chapter 8

Data Structure Documentation

8.1 Flexio_Pwm_Ip_ChannelConfigType Struct Reference

PWM configuration parameters structure.

`#include <Flexio_Pwm_Ip_Types.h>`

Data Fields

- uint8 TimerId
  Flexio used timer index.
- uint8 PinId
  Flexio used pin index.
- uint16 Period
  Pwm period in ticks.
- uint16 DutyCycle
  Pwm duty cycle in ticks.
- Flexio_Pwm_Ip_InterruptType IrqMode
  Interrupt mode.
- Flexio_Pwm_Ip_IplNotificationType IplCallback
  User notification callback for IPL.
- Flexio_Pwm_Ip_HldNotificationType HldCallback
  User notification callback for HLD.

8.1.1 Detailed Description PWM configuration parameters structure.

Flexio IP specific channel configuration structure type

Definition at line 250 of file Flexio_Pwm_Ip_Types.h.

8.1.2 Field Documentation

8.1.2.1 TimerId

`uint8 TimerId`

Flexio used timer index.

Definition at line 253 of file Flexio_Pwm_Ip_Types.h.

8.1.2.2 PinId

`uint8 PinId`

Flexio used pin index.

Definition at line 255 of file Flexio_Pwm_Ip_Types.h.

8.1.2.3 Period

`uint16 Period`

Pwm period in ticks.

Definition at line 261 of file Flexio_Pwm_Ip_Types.h.

8.1.2.4 DutyCycle

`uint16 DutyCycle`

Pwm duty cycle in ticks.

Definition at line 263 of file Flexio_Pwm_Ip_Types.h.

8.1.2.5 IrqMode

`Flexio_Pwm_Ip_InterruptType IrqMode`

Interrupt mode.

Definition at line 269 of file Flexio_Pwm_Ip_Types.h.

8.1.2.6 IplCallback

`Flexio_Pwm_Ip_IplNotificationType IplCallback`

User notification callback for IPL.

Definition at line 271 of file Flexio_Pwm_Ip_Types.h.

8.1.2.7 HldCallback

`Flexio_Pwm_Ip_HldNotificationType HldCallback`

User notification callback for HLD.

Definition at line 273 of file Flexio_Pwm_Ip_Types.h.

8.2 Flexio_Pwm_Ip_HldNotificationType Struct Reference

Structure for notification.

`#include <Flexio_Pwm_Ip_Types.h>`

Data Fields

- Flexio_Pwm_Ip_HldCallbackType CbFunction
  Callback function pointer.
- uint8 CbParameter
  Callback function parameter pointer.

8.2.1 Detailed Description

Structure for notification.

The structure used to notification

Definition at line 236 of file Flexio_Pwm_Ip_Types.h.

8.2.2 Field Documentation

8.2.2.1 CbFunction

`Flexio_Pwm_Ip_HldCallbackType CbFunction`

Callback function pointer.

Definition at line 239 of file Flexio_Pwm_Ip_Types.h.

8.2.2.2 CbParameter

`uint8 CbParameter`

Callback function parameter pointer.

Definition at line 241 of file Flexio_Pwm_Ip_Types.h.

8.3 Flexio_Pwm_Ip_IplNotificationType Struct Reference

Structure for notification.

`#include <Flexio_Pwm_Ip_Types.h>`

Data Fields

- Flexio_Pwm_Ip_IplCallbackType CbFunction
  Callback function pointer.
- void * CbParameter
  Callback function parameter pointer.

8.3.1 Detailed Description

Structure for notification.

The structure used to notification

Definition at line 222 of file Flexio_Pwm_Ip_Types.h.

8.3.2 Field Documentation

8.3.2.1 CbFunction

`Flexio_Pwm_Ip_IplCallbackType CbFunction`

Callback function pointer.

Definition at line 225 of file Flexio_Pwm_Ip_Types.h.

8.3.2.2 CbParameter

`void* CbParameter`

Callback function parameter pointer.

Definition at line 227 of file Flexio_Pwm_Ip_Types.h.

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